

Chapter 6

Trust-Region Methods and Conic Model Methods

6.1 Trust-Region Methods

6.1.1 Trust-Region Methods

The basic idea of Newton's method is to approximate the objective function $f(x)$ around x_k by choosing a quadratic model of the form

$$q^{(k)}(s) = f(x_k) + g_k^T s + \frac{1}{2} s^T G_k s,$$

where $g_k = \nabla f(x_k)$ and $G_k = \nabla^2 f(x_k)$, and use the minimizer s_k of $q^{(k)}(s)$ to modify x_k ,

$$x_{k+1} = x_k + s_k.$$

However, this method can only guarantee the local convergence, i.e., when s is small enough, the method is convergent locally. In Chapter 2, we have introduced line search approaches which guarantee the method is convergent globally. Line search approaches use the quadratic model to generate a search direction and then find a suitable stepsize α along the direction. Although it is successful at most time, it does not use sufficiently the n -dimensional quadratic model. The other disadvantage is that the Newton's method cannot be used if the Hessian matrices are not positive definite.

In this section the other class of global approaches is introduced, which is called the trust-region method. It not only replaces line search to get

the global convergence, but also circumvents the difficulty caused by non-positive definite Hessian matrices in line search. Besides, it produces more significant reduction in objective value f than line search approaches. In the trust-region method, we first define a region around the current iterate

$$\Omega_k = \{x : \|x - x_k\| \leq \Delta_k\},$$

where Δ_k is the radius of Ω_k , in which the model is trusted to be adequate to the objective function. And then we choose a step to be the approximate minimizer of the quadratic model in the trust-region, i.e., such that $x_k + s_k$ is the approximately best point on the generalized sphere

$$\{x_k + s \mid \|s\| \leq \Delta_k\}$$

with center x_k and radius Δ_k . If the step is not acceptable, we reduce the size of the trust-region and find a new minimizer. This method retains the rapid local convergence rate of Newton's method and quasi-Newton method, but also has ideal global convergence. Since the step is restricted by the trust-region, it is also called the restricted step method. The model subproblem of the trust-region method is

$$\begin{aligned} \min \quad & q^{(k)}(s) = f(x_k) + g_k^T s + \frac{1}{2} s^T B_k s \\ \text{s.t.} \quad & \|s\| \leq \Delta_k, \end{aligned} \tag{6.1.1}$$

where $\Delta_k > 0$ is the trust-region radius, B_k is symmetric and approximate to the Hessian G_k . Normally, we use l_2 norm $\|\cdot\|_2$ so that s_k is the minimizer of $q^{(k)}(s)$ in the ball of radius Δ_k . Other norms can also be used, however, the different norms define the different shapes of the trust-region. In (6.1.1), if we set $B_k = G_k$, the method is said to be a Newton-type trust-region method.

How to choose Δ_k at each iteration? In general, when there is good agreement between the model $q^{(k)}(s)$ and the objective function value $f(x_k + s)$, one should select Δ_k as large as possible. Let

$$Ared_k = f(x_k) - f(x_k + s_k) \tag{6.1.2}$$

which is called the *actual reduction*, and let

$$Pred_k = q^{(k)}(0) - q^{(k)}(s_k) \tag{6.1.3}$$

which is called the *predicted reduction*. Define the ratio

$$r_k = \frac{Ared_k}{Pred_k}, \quad (6.1.4)$$

which measures the agreement between the model function $q^{(k)}$ and the objective function f . This ratio r_k plays an important role in selecting new iterate x_{k+1} and updating the trust-region radius Δ_k . If r_k is close to 1, it means there is good agreement, and we can expand the trust-region for the next iteration; if r_k is close to zero or negative, we shrink the trust-region; otherwise, we do not alter the trust-region. The following is the trust-region algorithm.

Algorithm 6.1.1 (*Trust-Region Algorithm*)

Step 1. Given initial point $x_0, \bar{\Delta}, \Delta_0 \in (0, \bar{\Delta}), \epsilon \geq 0, 0 < \eta_1 \leq \eta_2 < 1$ and $0 < \gamma_1 < 1 < \gamma_2, k := 0$.

Step 2. If $\|g_k\| \leq \epsilon$, stop.

Step 3. Approximately solve the subproblem (6.1.1) for s_k .

Step 4. Compute $f(x_k + s_k)$ and r_k . Set

$$x_{k+1} = \begin{cases} x_k + s_k, & \text{if } r_k \geq \eta_1, \\ x_k, & \text{otherwise.} \end{cases}$$

Step 5. If $r_k < \eta_1$, then $\Delta_{k+1} \in (0, \gamma_1 \Delta_k]$;

If $r_k \in [\eta_1, \eta_2)$, then $\Delta_{k+1} \in [\gamma_1 \Delta_k, \Delta_k]$;

If $r_k \geq \eta_2$ and $\|s_k\| = \Delta_k$, then $\Delta_{k+1} \in [\Delta_k, \min\{\gamma_2 \Delta_k, \bar{\Delta}\}]$.

Step 6. Generate B_{k+1} , update $q^{(k)}$, set $k := k + 1$, go to Step 2.

□

In the above algorithm, $\bar{\Delta}$ is an overall bound for all Δ_k . The iterations for which $r_k \geq \eta_2$ and thus for which $\Delta_{k+1} \geq \Delta_k$, are said to be very successful iterations; the iterations for which $r_k \geq \eta_1$ and thus for which $x_{k+1} = x_k + s_k$, are said to be successful iterations; otherwise the iterations for which $r_k < \eta_1$ and thus for which $x_{k+1} = x_k$, are said to be unsuccessful iterations. Sometimes, the iterations in the first two cases are said to be successful iterations.

We like to point out some choices of the parameters, for instance, $\eta_1 = 0.01, \eta_2 = 0.75, \gamma_1 = 0.5, \gamma_2 = 2, \Delta_0 = 1$ or $\Delta_0 = \frac{1}{10}\|g_0\|$. However, the algorithm is insensitive to their change. In addition, Δ_{k+1} can be selected by polynomial interpolation. For example, if $r_k < 0.01$, then Δ_{k+1} can be chosen in an interval $(0.01, 0.5)\|s_k\|$ on the basis of a polynomial interpolation. Also, if we use quadratic interpolation, we have

$$\lambda = \frac{-g_k^T s_k}{2[f(x_k + s_k) - f(x_k) - g_k^T s_k]}, \quad (6.1.5)$$

and we set

$$\Delta_{k+1} = \lambda\|s_k\|. \quad (6.1.6)$$

Finally, to conclude this subsection, we give the characterization of the solution of subproblem (6.1.1). For convenience, we drop the subscripts in the following theorem.

Theorem 6.1.2 *The vector s^* is the solution of the subproblem*

$$\min \quad f + g^T s + \frac{1}{2}s^T B s \quad (6.1.7)$$

$$\text{s.t.} \quad \|s\|_2 \leq \Delta, \quad (6.1.8)$$

if and only if there is a scalar $\lambda^ \geq 0$ such that*

$$(B + \lambda^* I)s^* = -g, \quad (6.1.9)$$

$$\|s^*\|_2 \leq \Delta, \quad (6.1.10)$$

$$\lambda^*(\Delta - \|s^*\|_2) = 0, \quad (6.1.11)$$

and $(B + \lambda^ I)$ is positive semidefinite.*

Proof. Let s^* be the solution of subproblem (6.1.7)-(6.1.8). From the optimality condition of constrained optimization (see Chapter 8), there exists a multiplier $\lambda^* \geq 0$ such that (6.1.9)-(6.1.11) hold. We now need to prove that the matrix $(B + \lambda^* I)$ is positive semidefinite.

If $\|s^*\|_2 < \Delta$, then $\lambda^* = 0$ and s^* is an unconstrained minimizer of q , and thus B is positive semidefinite and furthermore $(B + \lambda^* I)$ is positive semidefinite.

If $\|s^*\|_2 = \Delta$, it follows from the second-order necessary condition (see §8.3) that

$$s^T (B + \lambda^* I) s \geq 0 \quad (6.1.12)$$

for all s satisfying $s^T s^* = 0$. If $s^T s^* \neq 0$, take $t = -2s^T s^* / \|s\|_2^2$, then $\|s^* + ts\|_2 = \Delta$. By the definition of s^* , we have

$$q(s^* + ts) + \frac{1}{2}\lambda^* \|s^* + ts\|_2^2 \geq q(s^*) + \frac{1}{2}\lambda^* \|s^*\|_2^2. \quad (6.1.13)$$

Developing $q(\cdot)$ yields

$$\frac{1}{2}t^2 s^T (B + \lambda^* I) s \geq -t(s^T [g + (B + \lambda^* I)s^*]). \quad (6.1.14)$$

By using (6.1.9) we get that the right-hand side of (6.1.14) is equal to zero. Then the above inequality indicates that

$$s^T (B + \lambda^* I) s \geq 0$$

for all s with $s^T s^* \neq 0$. Therefore, $B + \lambda^* I$ is positive semidefinite.

Conversely, assume that there is $\lambda^* \geq 0$ such that s^* satisfies (6.1.9)-(6.1.11) and that $B + \lambda^* I$ is positive semidefinite. Then, for all s satisfying $\|s\|_2 \leq \Delta$, we have

$$\begin{aligned} q(s) &= f(x) + g^T s + \frac{1}{2}s^T (B + \lambda^* I) s - \frac{1}{2}\lambda^* \|s\|_2^2 \\ &\geq f(x) + g^T s^* + \frac{1}{2}(s^*)^T (B + \lambda^* I) s^* - \frac{1}{2}\lambda^* \|s\|_2^2 \\ &= q(s^*) + \frac{1}{2}\lambda^* [\|s^*\|_2^2 - \|s\|_2^2]. \end{aligned}$$

By use of (6.1.11), we have that $\lambda^*(\Delta^2 - (s^*)^T s^*) = 0$. So, the above inequality becomes

$$\begin{aligned} q(s) &\geq q(s^*) + \frac{1}{2}\lambda^* [(\|s^*\|_2^2 - \Delta^2) + (\Delta^2 - \|s\|_2^2)] \\ &= q(s^*) + \frac{1}{2}\lambda^* [\Delta^2 - \|s\|_2^2]. \end{aligned}$$

Thus, from $\lambda^* \geq 0$ and $\|s\|_2 \leq \Delta$, we immediately have

$$q(s) \geq q(s^*),$$

which implies s^* is the solution of (6.1.7)-(6.1.8). $\square$

If $(B + \lambda^* I)$ is singular, we refer to this case as the *hard case*. In this case, d^* has the form

$$d^* = -(B + \lambda^* I)^+ g + v, \quad (6.1.15)$$

where $(B + \lambda^* I)^+$ denotes the generalized inverse of $(B + \lambda^* I)$, and v is a vector in null space of $(B + \lambda^* I)$.

Assume that $(B + \lambda^* I)$ is positive definite, then d^* can be obtained by solving

$$\lambda[\Delta - \|(B + \lambda I)^{-1}g\|_2] = 0, \quad (6.1.16)$$

$$\|(B + \lambda I)^{-1}g\|_2 \leq \Delta, \quad \lambda \geq 0 \quad (6.1.17)$$

for λ^* and then setting

$$d^* = -(B + \lambda^* I)^{-1}g.$$

If B is positive definite and $\|B^{-1}g\|_2 < \Delta$, then $d^* = -B^{-1}g$ simply. Otherwise, $\lambda^* > 0$. We need to solve

$$\psi(\lambda) = \frac{1}{\|(B + \lambda I)^{-1}g\|_2} - \frac{1}{\Delta} = 0. \quad (6.1.18)$$

We consider solving $\psi(\lambda) = 0$, instead of solving $\Delta - \|(B + \lambda I)^{-1}g\|_2 = 0$, because $\psi(\lambda)$ is nearly linear in the considered range. By direct computation, we have

$$\psi'(\lambda) = \frac{g^T H(\lambda)^{-3}g}{\|H(\lambda)^{-1}g\|_2^3}, \quad (6.1.19)$$

$$\psi''(\lambda) = -\frac{3g^T H(\lambda)^{-4}g}{\|H(\lambda)^{-1}g\|_2^3} [1 - \cos^2(\langle H(\lambda)^{-1}g, H(\lambda)^{-2}g \rangle)], \quad (6.1.20)$$

where $H(\lambda) = B + \lambda I$. Therefore, for the most negative eigenvalue $\lambda_1 < 0$, if $\lambda > -\lambda_1$, $\psi(\lambda)$ is strictly increasing and concave. So, Newton's method can be used to solve (6.1.18), that is,

$$\begin{aligned} \lambda_+ &= \lambda - \frac{\psi(\lambda)}{\psi'(\lambda)} \\ &= \lambda - \frac{1}{\frac{g^T(B+\lambda I)^{-3}g}{\|(B+\lambda I)^{-1}g\|_2^3}} \left[\frac{1}{\|(B + \lambda I)^{-1}g\|_2} - \frac{1}{\Delta} \right]. \end{aligned} \quad (6.1.21)$$

6.1.2 Convergence of Trust-Region Methods

In order to discuss the convergence of trust-region methods, we first give some assumptions and technical lemmas.

We assume that the approximate Hessians B_k are uniformly bounded in norm, and that the level set

$$\{x \mid f(x) \leq f(x_0)\} \quad (6.1.22)$$

is bounded, on which the function $f : R^n \rightarrow R$ is continuously differentiable. For generality, we also allow the length of the approximate solution s_k of the subproblem (6.1.1) to exceed the trust-region bound, provided that it stays within a fixed multiple of the bound, that is

$$\|s_k\| \leq \tilde{\eta} \Delta_k, \quad (6.1.23)$$

where $\tilde{\eta}$ is a positive constant. The above assumptions are said to be Assumption (A_0) .

For trust-region algorithm, in general, we do not seek an accurate solution of subproblem (6.1.1) but we are content with a nearly optimal solution of (6.1.1). Strong theoretical and numerical results can be obtained if the step s_k produced by Algorithm 6.1.1 satisfies

$$q_k(0) - q_k(s_k) \geq \beta_1 \|g_k\|_2 \min \left\{ \Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2} \right\}, \quad (6.1.24)$$

where $\beta_1 \in (0, 1]$. Below, we show that the Cauchy point s_k^c satisfies (6.1.24) with $\beta_1 = \frac{1}{2}$ and that the exact solution s_k of the subproblem (6.1.1) satisfies (6.1.24) with $\beta_1 = \frac{1}{2}$. If s_k is an approximate solution of the subproblem (6.1.1) with $q^{(k)}(0) - q^{(k)}(s_k) \geq \beta_2 (q^{(k)}(0) - q^{(k)}(s_k^c))$, then it satisfies (6.1.24) with $\beta_1 = \frac{1}{2} \beta_2$.

Lemma 6.1.3 *Let s_k be the solution of (6.1.1), let $\|\cdot\| = \|\cdot\|_2$, then*

$$\begin{aligned} Pred_k &= q^{(k)}(0) - q^{(k)}(s_k) \\ &\geq \frac{1}{2} \|g_k\|_2 \min \left\{ \Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2} \right\}. \end{aligned} \quad (6.1.25)$$

Proof. By the definition of s_k , for all $\alpha \in [0, 1]$, we have

$$\begin{aligned} q^{(k)}(0) - q^{(k)}(s_k) &\geq q^{(k)}(0) - q^{(k)}\left(-\alpha \frac{\Delta_k}{\|g_k\|_2} g_k\right) \\ &= \alpha \Delta_k \|g_k\|_2 - \frac{1}{2} \alpha^2 \Delta_k^2 g_k^T B_k g_k / \|g_k\|_2^2 \\ &\geq \alpha \Delta_k \|g_k\|_2 - \frac{1}{2} \alpha^2 \Delta_k^2 \|B_k\|_2. \end{aligned} \quad (6.1.26)$$

Therefore we must have

$$\begin{aligned} Pred_k &\geq \max_{0 \leq \alpha \leq 1} [\alpha \Delta_k \|g_k\|_2 - \frac{1}{2} \alpha^2 \Delta_k^2 \|B_k\|_2] \\ &\geq \frac{1}{2} \|g_k\|_2 \min \left\{ \Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2} \right\}. \quad \square \end{aligned} \quad (6.1.27)$$

The Cauchy point of the subproblem (6.1.1) can be defined by

$$q^{(k)}(s_k^c) = \min\{q^{(k)}(s) \mid s = \tau s_k^G, \|s\| \leq \Delta_k\}, \quad (6.1.28)$$

where s_k^G solves a linear version of subproblem (6.1.1):

$$\begin{aligned} \min \quad & f(x_k) + g_k^T s \\ \text{s.t.} \quad & \|s\| \leq \Delta_k. \end{aligned} \quad (6.1.29)$$

Obviously, the solution of (6.1.29) is

$$s_k^G = -\frac{\Delta_k}{\|g_k\|_2} g_k.$$

Therefore, the Cauchy point of the subproblem (6.1.1) can be expressed as

$$s_k^c = \tau_k s_k^G = -\tau_k \frac{\Delta_k}{\|g_k\|_2} g_k, \quad (6.1.30)$$

where

$$\tau_k = \begin{cases} 1 & \text{if } g_k^T B_k g_k \leq 0; \\ \min\{\|g_k\|_2^3 / (\Delta_k g_k^T B_k g_k), 1\} & \text{otherwise.} \end{cases} \quad (6.1.31)$$

In fact, if $g_k^T B_k g_k \leq 0$, the function $q^{(k)}(s_k^c) = q^{(k)}(\tau s_k^G)$ decreases monotonically with τ when $g_k \neq 0$. Therefore, we can take τ as large as possible within $\|\tau s_k^G\| \leq \Delta_k$. In this case, by use of (6.1.30) and $\|\tau s_k^G\| \leq \Delta_k$, we have that $\tau_k = 1$. If $g_k^T B_k g_k > 0$, $q^{(k)}(\tau s_k^G)$ is a convex and quadratic function in τ . Then, by minimizing $q^{(k)}(\tau s_k^G)$, we obtain that τ_k equals $\|g_k\|_2^3 / (\Delta_k g_k^T B_k g_k)$, or the boundary value 1.

Lemma 6.1.4 *The Cauchy point s_k^c satisfies*

$$q^{(k)}(0) - q^{(k)}(s_k^c) \geq \frac{1}{2} \|g_k\|_2 \min \left\{ \Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2} \right\}. \quad (6.1.32)$$

Proof. Consider first the case of $g_k^T B_k g_k \leq 0$. In this case, it follows from (6.1.31) that $\tau_k = 1$, and we have

$$\begin{aligned}
 q^{(k)}(0) - q^{(k)}(s_k^c) &= -q^{(k)}\left(-\frac{\Delta_k}{\|g_k\|_2}g_k\right) \\
 &= \Delta_k \|g_k\|_2 - \frac{1}{2}\Delta_k^2 g_k^T B_k g_k / \|g_k\|_2^2 \\
 &\geq \Delta_k \|g_k\|_2 \\
 &\geq \|g_k\|_2 \min\left\{\Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2}\right\}.
 \end{aligned} \tag{6.1.33}$$

Consider the case of $g_k^T B_k g_k > 0$ and

$$\frac{\|g_k\|_2^3}{\Delta_k g_k^T B_k g_k} \leq 1. \tag{6.1.34}$$

In this case, $\tau_k = \|g_k\|^3 / (\Delta_k g_k^T B_k g_k)$, and we have

$$\begin{aligned}
 q^{(k)}(0) - q^{(k)}(s_k^c) &= \frac{\|g_k\|_2^4}{g_k^T B_k g_k} - \frac{1}{2}g_k^T B_k g_k \frac{\|g_k\|_2^4}{(g_k^T B_k g_k)^2} \\
 &= \frac{1}{2} \frac{\|g_k\|_2^4}{g_k^T B_k g_k} \\
 &\geq \frac{1}{2} \frac{\|g_k\|_2^2}{\|B_k\|_2} \\
 &\geq \frac{1}{2} \|g_k\|_2 \min\left\{\Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2}\right\}.
 \end{aligned} \tag{6.1.35}$$

Consider the case of $g_k^T B_k g_k > 0$ and

$$\frac{\|g_k\|_2^3}{\Delta_k g_k^T B_k g_k} > 1. \tag{6.1.36}$$

In this case, $\tau_k = 1$, and by use of (6.1.36) we have

$$\begin{aligned}
 q^{(k)}(0) - q^{(k)}(s_k^c) &= \Delta_k \|g_k\|_2 - \frac{1}{2}\Delta_k^2 g_k^T B_k g_k / \|g_k\|_2^2 \\
 &\geq \Delta_k \|g_k\|_2 - \frac{1}{2} \frac{\Delta_k^2}{\|g_k\|_2^2} \frac{\|g_k\|_2^3}{\Delta_k} \\
 &= \frac{1}{2} \Delta_k \|g_k\|_2 \\
 &\geq \frac{1}{2} \|g_k\|_2 \min\left\{\Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2}\right\}.
 \end{aligned} \tag{6.1.37}$$

The above discussion of three cases gives the result (6.1.32). $\square$

Usually, we assume that s_k is an approximate solution of the subproblem (6.1.1) and satisfies

$$q^{(k)}(0) - q^{(k)}(s_k) \geq \beta_2(q^{(k)}(0) - q^{(k)}(s_k^e)), \quad (6.1.38)$$

where s_k^e is an exact solution of subproblem (6.1.1) and $\beta_2 \in (0, 1]$ is a constant. Since $q^{(k)}(s_k^e) \leq q^{(k)}(s_k^c)$, we immediately have

$$q^{(k)}(0) - q^{(k)}(s_k) \geq \beta_2(q^{(k)}(0) - q^{(k)}(s_k^c)), \quad (6.1.39)$$

where $s_k^c = -\tau_k \frac{\Delta_k}{\|g_k\|_2} g_k$ with $0 \leq \tau_k \leq 1$ is a Cauchy point. So, we immediately have

Lemma 6.1.5 *Let s_k be an approximate solution of (6.1.1) and satisfy (6.1.38) or (6.1.39). Then*

$$\begin{aligned} \text{Pred}_k &= q^{(k)}(0) - q^{(k)}(s_k) \\ &\geq \frac{1}{2}\beta_2\|g_k\|_2 \min \left\{ \Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2} \right\}, \end{aligned} \quad (6.1.40)$$

where $\beta_2 \in (0, 1]$.

Next, in order to prove the global convergence theorem, we give some technical lemmas.

Lemma 6.1.6 *Let Assumption (A_0) hold. We have*

$$|f(x_k + s_k) - q^{(k)}(s_k)| \leq \frac{1}{2}M\|s_k\|^2 + C(\|s_k\|)\|s_k\|, \quad (6.1.41)$$

where $C(\|s_k\|)$ is arbitrarily small by restricting the size of s_k .

Proof. By Taylor's theorem,

$$f(x_k + s_k) = f(x_k) + g_k^T s_k + \int_0^1 [\nabla f(x_k + ts_k) - \nabla f(x_k)]^T s_k dt.$$

Also,

$$q^{(k)}(s_k) = f(x_k) + g_k^T s_k + \frac{1}{2}s_k^T B_k s_k.$$

Then

$$\begin{aligned} |f(x_k + s_k) - q^{(k)}(s_k)| &= \left| \frac{1}{2}s_k^T B_k s_k - \int_0^1 [\nabla f(x_k + ts_k) - \nabla f(x_k)]^T s_k dt \right| \\ &\leq \frac{1}{2}M\|s_k\|^2 + C(\|s_k\|)\|s_k\|. \quad \square \end{aligned}$$

Lemma 6.1.7 *Assume that Assumption (A_0) holds. Suppose that $\|g_k\|_2 \geq \epsilon > 0$ and that Δ_k is smaller than some threshold $\tilde{\Delta}$. Then the k -th iteration is a very successful iteration which satisfies $\Delta_{k+1} \geq \Delta_k$.*

Proof. By Lemma 6.1.5 and the assumptions,

$$\begin{aligned} \text{Pred}_k &= q^{(k)}(0) - q^{(k)}(s_k) \\ &\geq \frac{1}{2}\beta_2\|g_k\|_2 \min\left\{\Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2}\right\} \\ &\geq \frac{1}{2}\beta_2\epsilon \min\left\{\Delta_k, \frac{\epsilon}{M}\right\}. \end{aligned} \quad (6.1.42)$$

From Algorithm 6.1.1, by use of (6.1.41), (6.1.42) and (6.1.23), we have

$$\begin{aligned} |r_k - 1| &= \left| \frac{(f(x_k) - f(x_k + s_k)) - (q^{(k)}(0) - q^{(k)}(s_k))}{q^{(k)}(0) - q^{(k)}(s_k)} \right| \\ &= \left| \frac{f(x_k + s_k) - q^{(k)}(s_k)}{q^{(k)}(0) - q^{(k)}(s_k)} \right| \\ &\leq \frac{\frac{1}{2}M\|s_k\|^2 + C(\|s_k\|)\|s_k\|}{\frac{1}{2}\beta_2\epsilon \min\{\Delta_k, \epsilon/M\}} \\ &\leq \frac{\tilde{\eta}\Delta_k(M\tilde{\eta}\Delta_k + 2C(\|s_k\|))}{\beta_2\epsilon \min\{\Delta_k, \epsilon/M\}}. \end{aligned} \quad (6.1.43)$$

Since Δ_k is smaller than some threshold $\tilde{\Delta}$, we may choose $\tilde{\Delta}$ to be small enough such that

$$\Delta_k \leq \tilde{\Delta} \leq \epsilon/M, \quad M\tilde{\eta}\Delta_k + 2C(\|s_k\|) \leq (1 - \eta_2)\beta_2\epsilon/\tilde{\eta},$$

so we have $r_k \geq \eta_2$. It follows from Algorithm 6.1.1 that $\Delta_{k+1} \geq \Delta_k$. $\square$

This lemma indicates that if the current iterate is not a first-order stationary point and the trust-region radius Δ_k is small enough, then we always have $\Delta_{k+1} \geq \Delta_k$ and the iteration is very successful. Now we are in a position to give the global convergence theorem.

First, we consider the case when there are only finitely many successful iterations.

Theorem 6.1.8 *Under Assumption (A_0) , if Algorithm 6.1.1 has finitely many successful iterations, then the algorithm converges to the first-order stationary point.*

Proof. Since the algorithm has only finitely many successful iterations, then, for sufficiently large k , the iteration is unsuccessful. Thus, the sequence $\{\Delta_k\}$ from the algorithm converges to zero.

Suppose that k_0 is the index of the last successful iteration. If $\|g_{k_0+1}\| > 0$, it follows from Lemma 6.1.7 that there must be a very successful iteration of index larger than k_0 , which satisfies $\Delta_{k_0+1+1} > \Delta_{k_0+1}$. This is a contradiction to the assumption. The contradiction proves our theorem. $\square$

Next, we only need to restrict our attention to the case where there are infinitely many successful iterations.

Theorem 6.1.9 *Let Assumption (A_0) hold. If Algorithm 6.1.1 has infinitely many successful iterations, then the sequence of Algorithm 6.1.1 satisfies*

$$\liminf_{k \rightarrow \infty} \|g_k\| = 0. \quad (6.1.44)$$

Proof. Assume, by contradiction, that there is $\epsilon > 0$ and a positive index K such that

$$\|g_k\| \geq \epsilon \text{ for all } k \geq K.$$

From Algorithm 6.1.1 and Lemma 6.1.5, it follows for successful iterations that

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geq \eta_1 [q^{(k)}(0) - q^{(k)}(s_k)] \\ &\geq \frac{1}{2} \eta_1 \beta_2 \|g_k\|_2 \min \left[\Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2} \right] \\ &\geq \frac{1}{2} \eta_1 \beta_2 \epsilon \min \left[\Delta_k, \frac{\epsilon}{\beta} \right], \end{aligned} \quad (6.1.45)$$

where $\beta = \max\{1 + \|B_k\|_2\}$ is an upper bound of the Hessian approximation. So,

$$\begin{aligned} f(x_0) - f(x_{k+1}) &= \sum_{j=0, j \in \mathcal{S}} [f(x_j) - f(x_{j+1})] \\ &\geq \frac{1}{2} \sigma_k \eta_1 \beta_2 \epsilon \min \left[\Delta_k, \frac{\epsilon}{\beta} \right], \end{aligned}$$

where σ_k is a number of successful iterations till the k -th iteration with

$$\lim_{k \rightarrow \infty} \sigma_k = +\infty,$$

and $\mathcal{S}$ is an index set of successful iterations.

Since f is bounded below, it follows from the above inequality that

$$\lim_{k \rightarrow \infty} \Delta_k = 0, \quad (6.1.46)$$

contradicting the conclusion of Lemma 6.1.7. $\square$

Now we give a stronger result on the convergence which is for all limit points.

Theorem 6.1.10 *Suppose that Assumption (A_0) holds. Then*

$$\lim_{k \rightarrow \infty} g_k = 0. \quad (6.1.47)$$

Proof. Assume, by contradiction, that the conclusion does not hold, then there is a subsequence of successful iterations such that

$$\|g_{t_i}\| \geq 2\epsilon > 0 \quad (6.1.48)$$

for some $\epsilon > 0$ and for all i .

Theorem 6.1.9 guarantees that, for each i , there exists a first successful iteration $l(t_i) > t_i$ such that $\|g_{l(t_i)}\| < \epsilon$. We denote $l_i \triangleq l(t_i)$. Thus, there exists another subsequence $\{l_i\}$ such that

$$\|g_k\| \geq \epsilon \quad \text{for} \quad t_i \leq k < l_i \quad \text{and} \quad \|g_{l_i}\| < \epsilon. \quad (6.1.49)$$

Since

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geq \eta_1[q^{(k)}(0) - q^{(k)}(s_k)] \\ &\geq \frac{1}{2}\eta_1\beta_2\epsilon \min[\Delta_k, \epsilon/\beta], \end{aligned} \quad (6.1.50)$$

it follows from the monotonically decreasing and the bounded below of the sequence $\{f(x_k)\}$ that

$$\lim_{k \rightarrow \infty} \Delta_k = 0. \quad (6.1.51)$$

Then

$$\Delta_k \leq \frac{2}{\eta_1\beta_2\epsilon}[f(x_k) - f(x_{k+1})] \quad (6.1.52)$$

which implies that for i sufficiently large,

$$\begin{aligned} \|x_{t_i} - x_{l_i}\| &\leq \sum_{j=t_i}^{l_i-1} \|x_j - x_{j+1}\| \leq \sum_{j=t_i}^{l_i-1} \Delta_j \\ &\leq \frac{2}{\eta_1\beta_2\epsilon}[f(x_{t_i}) - f(x_{l_i})]. \end{aligned} \quad (6.1.53)$$

From the fact that the right-hand side converges to zero, we get

$$\|x_{t_i} - x_{l_i}\| \rightarrow 0, \text{ when } i \rightarrow \infty,$$

which deduces from continuity of gradient that

$$\|g_{t_i} - g_{l_i}\| \rightarrow 0,$$

which contradicts (6.1.49), because (6.1.49) implies that $\|g_{t_i} - g_{l_i}\| \geq \epsilon$. The contradiction proves our conclusion. $\square$

6.1.3 Solving A Trust-Region Subproblem

The Dogleg Method and The Double Dogleg Method

An efficient implementation to solve the trust-region subproblem is the so-called dogleg method which was presented by Powell [260]. To find an approximate solution of the subproblem (6.1.1), i.e., to find $x_{k+1} = x_k + s_k$ such that $\|s_k\| = \Delta_k$, Powell used a path consisting of two line segments to approximate s . The first line segment runs from the origin to the Cauchy point (a minimizer C.P. generated by the steepest descent method); the second line segment runs from the Cauchy point C.P. to the Newton point (the minimizer x_{k+1}^N generated by Newton method or quasi-Newton method). Let x_{k+1} be the intersection point of the path and the trust-region boundary. Obviously, $\|x_{k+1} - x_k\| = \Delta_k$. When the Newton step s_k^N satisfies $\|s_k^N\| \leq \Delta_k$, the new iterate x_{k+1} is just the Newton point, $x_{k+1} = x_{k+1}^N = x_k - B_k^{-1}g_k$.

Dennis and Mei [90] found that if the point generated by trust-region iteration is biased towards the Newton direction, the behavior of the algorithm will be further improved. Then we choose a point $\hat{N}$ on the Newton direction, and connect the Cauchy point C.P. to $\hat{N}$. The intersection point of the connection line and the trust-region boundary is taken as the new iterate x_{k+1} (see $x_{k+1}^{(2)}$ in Figure 6.1.1). Comparatively, $x_{k+1}^{(2)}$ is more biased to the Newton direction than $x_{k+1}^{(1)}$. We say $x_k \rightarrow C.P. \rightarrow x_{k+1}^N$ as dogleg, and $x_k \rightarrow C.P. \rightarrow \hat{N} \rightarrow x_{k+1}^N$ as double dogleg.

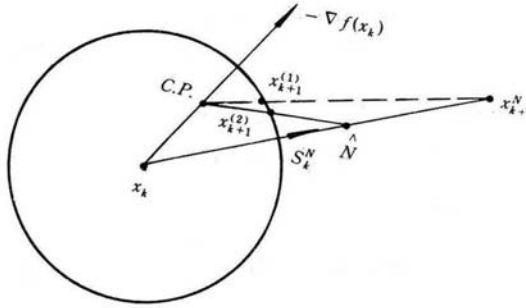


Figure 6.1.1 Dogleg method and double dogleg method

For quadratic model

$$q^{(k)}(x_k - \alpha g_k) = f(x_k) - \alpha \|g_k\|_2^2 + \frac{1}{2} \alpha^2 g_k^T B_k g_k,$$

the exact line search factor α_k has the obvious representation

$$\alpha_k = \frac{\|g_k\|_2^2}{g_k^T B_k g_k}.$$

Then the step along the steepest descent direction is

$$s_k^c = -\alpha_k g_k = -\frac{g_k^T g_k}{g_k^T B_k g_k} g_k. \quad (6.1.54)$$

If $\|s_k^c\|_2 = \|\alpha_k g_k\|_2 \geq \Delta_k$, we take

$$s_k = -\frac{\Delta_k}{\|g_k\|_2} g_k \quad (6.1.55)$$

and

$$x_{k+1} = x_k - \frac{\Delta_k}{\|g_k\|_2} g_k \quad (6.1.56)$$

which lies at the intersection of the negative gradient and the trust-region boundary. If $\|s_k^c\|_2 < \Delta_k$ and $\|s_k^N\|_2 > \Delta_k$, we take

$$s_k(\lambda) = s_k^c + \lambda(s_k^N - s_k^c), \quad 0 \leq \lambda \leq 1,$$

and thus

$$x_{k+1} = x_k + s_k(\lambda) = x_k + s_k^c + \lambda(s_k^N - s_k^c), \quad 0 \leq \lambda \leq 1, \quad (6.1.57)$$

where the value λ is obtained by solving the equation

$$\|s_k^c + \lambda(s_k^N - s_k^c)\|_2 = \Delta_k.$$

Otherwise, take

$$s_k = s_k^N = -B_k^{-1}g_k. \quad (6.1.58)$$

Combining (6.1.56), (6.1.57) and (6.1.58) yields

$$x_{k+1} = \begin{cases} x_k - \frac{\Delta_k}{\|g_k\|_2}g_k, & \text{when } \|s_k^c\|_2 \geq \Delta_k, \\ x_k + s_k^c + \lambda(s_k^N - s_k^c), & \text{when } \|s_k^c\|_2 < \Delta_k \text{ and } \|s_k^N\|_2 > \Delta_k, \\ x_k - B_k^{-1}g_k, & \text{when } \|s_k^c\|_2 < \Delta_k \text{ and } \|s_k^N\|_2 \leq \Delta_k, \end{cases} \quad (6.1.59)$$

where $0 \leq \lambda \leq 1$.

The following theorem demonstrates the property possessed by the dogleg method and the double dogleg method. In the following, as an example, we only consider the double dogleg method.

Theorem 6.1.11 *In the double dogleg method,*

1. *The distance from x_k to C.P., to $\hat{N}$, is increasing monotonically.*
2. *The model value $q^{(k)}(x_k + s)$ is decreasing monotonically when the point moves from x_k to C.P., to $\hat{N}$, and to x_{k+1}^N .*

Proof. (1) Since

$$\begin{aligned} \|s_k^c\| &= \|\alpha_k g_k\| = \|g_k\|_2^3 / g_k^T B_k g_k \\ &\leq \frac{\|g_k\|_2^3}{g_k^T B_k g_k} \frac{\|g_k\|_2 \|B_k^{-1}g_k\|_2}{g_k^T B_k^{-1}g_k} \\ &= \frac{\|g_k\|_2^4}{(g_k^T B_k g_k)(g_k^T B_k^{-1}g_k)} \|s_k^N\|_2 \\ &\triangleq \gamma \|s_k^N\|_2, \end{aligned} \quad (6.1.60)$$

it follows from Kantorovich inequality (3.1.33) that $\gamma \leq 1$ and then

$$\|s_k^c\|_2 \leq \gamma \|s_k^N\|_2 \leq \|s_k^N\|_2. \quad (6.1.61)$$

Take $\hat{N}$ being

$$x^{\hat{N}} = x_k - \eta B_k^{-1}g_k = x_k + \eta s_k^N, \quad (6.1.62)$$

where

$$\gamma \leq \eta \leq 1. \quad (6.1.63)$$

Thus

$$\|x^c - x_k\|_2 \leq \|x^{\hat{N}} - x_k\|_2 \leq \|x_{k+1}^N - x_k\|_2 \quad (6.1.64)$$

which shows the property (1) holds.

(2) It is enough to prove that $q^{(k)}(x_k + s)$ decreases monotonically when the point moves from the point C.P. to the point $\hat{N}$. In fact,

$$x_{k+1}(\lambda) = x_k + s_k^c + \lambda(\eta s_k^N - s_k^c), \quad 0 \leq \lambda \leq 1. \quad (6.1.65)$$

The direction derivative of $q^{(k)}$ at $x_{k+1}(\lambda)$ is

$$\begin{aligned} & \nabla q^{(k)}(x_{k+1}(\lambda))^T (\eta s_k^N - s_k^c) \\ &= (g_k + B_k s_k^c)^T (\eta s_k^N - s_k^c) + \lambda (\eta s_k^N - s_k^c)^T B_k (\eta s_k^N - s_k^c). \end{aligned} \quad (6.1.66)$$

When B_k is positive definite, the right-hand side of (6.1.66) is a monotone increasing function of λ . Therefore, in order to make the above equality negative when $0 \leq \lambda \leq 1$, it is enough to ask the above equality to be negative when $\lambda = 1$, i.e.,

$$(g_k + B_k s_k^c)^T (\eta s_k^N - s_k^c) + \lambda (\eta s_k^N - s_k^c)^T B_k (\eta s_k^N - s_k^c) < 0.$$

Developing and using $B_k s_k^N = -g_k$, the above inequality is equivalent to

$$0 > (1 - \eta)(g_k^T (\eta s_k^N - s_k^c)) = (1 - \eta)(\gamma - \eta)(g_k^T B_k^{-1} g_k). \quad (6.1.67)$$

Obviously, it is satisfied when $\gamma < \eta < 1$. Therefore the second property holds. $\square$

In summary, the double dogleg method chooses the point $\hat{N}$ which is defined by

$$x_{k+1} = x_k + \eta s_k^N, \quad \eta \in [\gamma, 1]. \quad (6.1.68)$$

When $\eta = 1$, the point $\hat{N}$ is just the Newton point x_{k+1}^N and the double dogleg step is just the dogleg step. Generally, we take $\eta = 0.8\gamma + 0.2$.

After generating the points C.P. and $\hat{N}$, we find $x_{k+1}(\lambda)$ by (6.1.65), such that

$$\|s_k^c + \lambda(\eta s_k^N - s_k^c)\|_2^2 = \Delta_k^2, \quad (6.1.69)$$

which is a one-dimensional root-finding problem and can be solved by Newton's method. If $x_{k+1}(\lambda)$ obtained satisfies the descent requirement

$$f(x_{k+1}(\lambda)) \leq f(x_k) + \rho g_k^T(x_{k+1}(\lambda) - x_k), \quad \rho \in (0, \frac{1}{2}), \quad (6.1.70)$$

$x_{k+1}(\lambda)$ will be accepted as new iterate x_{k+1} , and the trust-region will be updated by Step 4 in Algorithm 6.1.1; if $x_{k+1}(\lambda)$ does not satisfy (6.1.70), then set $x_{k+1} := x_k$.

Steihaug-CG Method

The methods for solving the trust-region subproblem described above require the solution of a linear system. When the problem is large, the operation may be quite costly. Steihaug [322] proposed a technique based on a preconditioned and truncated conjugate gradient method and trust-region method, which solves the trust-region subproblem approximately. This method is usually called Steihaug-CG method. Since it was independently proposed by Toint [341], it is also called the Steihaug-Toint method.

Consider a scaled trust-region subproblem

$$\min \quad q(s) = g^T s + \frac{1}{2} s^T B s \quad (6.1.71)$$

$$\text{s.t.} \quad \|s\|_W \leq \Delta, \quad (6.1.72)$$

(we drop the subscripts here for simplicity) where W is a symmetric and positive definite matrix. Steihaug applied the preconditioned conjugate gradient method (PCG) to subproblem (6.1.71)–(6.1.72), and considered three possible termination rules. Firstly, if $d_k^T B d_k > 0$, the method corresponds to the convex interior solution. Secondly, if $d_k^T B d_k \leq 0$, we meet a direction of negative curvature. In this case, we move to the trust-region boundary along the line $s_k + \tau d_k$ with $\tau > 0$ so that $\|s_k + \tau d_k\|_W = \Delta$. Finally, if the solution lies outside the trust-region, we ask that the new point be on the boundary.

The Steihaug-CG algorithm for trust-region subproblem is as follows.

Algorithm 6.1.12 (Steihaug-CG Algorithm for TR Subproblem)

Step 0. Given $\varepsilon > 0$. Let $s_0 = 0, g_0 = g, v_0 = W^{-1}g_0, d_0 = -v_0$.

If $\|g_0\| < \varepsilon$, set $s = s_0$, stop;

For $j = 0, 1, \dots$, perform the following steps:

Step 1. If $d_j^T B d_j \leq 0$, compute $\tau > 0$ so that $\|s_j + \tau d_j\|_W = \Delta$,
 set $s = s_j + \tau d_j$,
 stop;
 End if

Step 2. Set $\alpha_j = g_j^T v_j / d_j^T B d_j$;
 Set $s_{j+1} = s_j + \alpha_j d_j$;
 If $\|s_{j+1}\|_W \geq \Delta$, compute $\tau > 0$ so that $\|s_j + \tau d_j\|_W = \Delta$,
 set $s = s_j + \tau d_j$,
 stop;
 End if

Step 3. Set $g_{j+1} = g_j + \alpha_j B d_j$;
 If $\|g_{j+1}\|_W < \varepsilon \|g_0\|_W$, set $s = s_{j+1}$, stop;
 End if

Step 4. Set
 $v_{j+1} = W^{-1} g_{j+1}$,
 $\beta_j = g_{j+1}^T v_{j+1} / g_j^T v_j$,
 $d_{j+1} = -v_{j+1} + \beta_j d_j$.

This method has some properties similar to the dogleg method. Next, we state these properties. In the proof of the property theorem, we need the following lemma which is easy.

Lemma 6.1.13 Assume $d_i^T B d_i \neq 0$, then we have

$$g_i^T d_j = -g_j^T v_j, \quad 0 \leq i \leq j, \quad (6.1.73)$$

$$d_i^T W d_j = \frac{g_j^T v_j}{g_i^T v_i} d_i^T W d_i, \quad 0 \leq i \leq j, \quad (6.1.74)$$

$$q(s_{i+1}) = q(s_i) - \frac{1}{2} \frac{(g_i^T v_i)^2}{d_i^T B d_i}. \quad (6.1.75)$$

Proof. We can use the explicit formula for the steplength α and iterative scheme of PCG to get the results. In fact, by $g_j^T d_{j-1} = 0$ and the conjugacy of d_j and d_{j-1} , we have

$$\begin{aligned} g_j^T v_j &= g_j^T (-d_j + \beta_{j-1} d_{j-1}) \\ &= -g_j^T d_j \end{aligned}$$

$$\begin{aligned}
&= -(g_{j-1} + \alpha_{j-1} B d_{j-1})^T d_j \\
&= -g_{j-1}^T d_j \\
&= \dots \\
&= -g_i^T d_j, \quad i \leq j
\end{aligned}$$

which shows (6.1.73).

Next, we prove (6.1.74).

$$\begin{aligned}
d_j^T W d_i &= (-v_j + \beta_{j-1} d_{j-1})^T W d_i \\
&= -v_j^T W d_i + \beta_{j-1} d_{j-1}^T W d_i \\
&= -g_j^T d_i + \beta_{j-1} d_{j-1}^T W d_i \\
&= \beta_{j-1} d_{j-1}^T W d_i.
\end{aligned}$$

By recurrence, we have

$$\begin{aligned}
d_j^T W d_i &= \beta_{j-1} \beta_{j-2} \dots \beta_i d_i^T W d_i \\
&= \frac{g_j^T v_j}{g_i^T v_i} d_i^T W d_i
\end{aligned}$$

that shows (6.1.74).

For (6.1.75), it is a direct consequence of (4.3.40). $\square$

Now we are in a position to state the properties of the Steihaug-CG algorithm.

Theorem 6.1.14 *Let $\|s_j\|$ be the iterates generated by PCG Algorithm 6.1.12. Then $q(s_j)$ in (6.1.71) is strictly decreasing, i.e.,*

$$q(s_{j+1}) < q(s_j). \quad (6.1.76)$$

Further, $\|s_j\|_W$ is strictly increasing:

$$0 = \|s_0\|_W < \dots < \|s_j\|_W < \|s_{j+1}\|_W < \dots < \|s\|_W \leq \Delta. \quad (6.1.77)$$

Proof. We first prove (6.1.76). From (6.1.75), $q(s_j)$ is strictly decreasing.

Consider the last iterate s . If $s = s_{j+1}$, then the result follows directly. From (6.1.73) we have that

$$g_j^T d_j = -g_j^T v_j = -(Bs_j + g)^T W^{-1}(Bs_j + g) < 0,$$

hence d_j is a descent direction for $q(s_j)$. If $d_j^T B d_j > 0$, then

$$q(s_j) \geq q(s_j + \tau d_j) \geq q(s_{j+1}), \text{ for } 0 < \tau \leq \alpha_j.$$

Since $\tau \leq \alpha_j$, we have the desired result.

For $d_j^T B d_j \leq 0$, then the quadratic term is non-positive, and we have

$$q(s_j) \geq q(s_j + \tau d_j), \text{ for } \tau \geq 0,$$

and the result follows.

Now we show that $\|s_j\|_W$ is strictly increasing and that (6.1.77) holds. From Algorithm 6.1.12, we have

$$s_j = s_0 + \sum_{k=0}^{j-1} \alpha_k d_k = \sum_{k=0}^{j-1} \alpha_k d_k \quad (6.1.78)$$

and

$$\alpha_k > 0, \quad k = 0, 1, \dots, j-1. \quad (6.1.79)$$

Hence, by (6.1.78) and (6.1.74), we have

$$s_j^T W d_j = \sum_{k=0}^{j-1} \alpha_k d_k^T W d_j > 0. \quad (6.1.80)$$

Using (6.1.80) and (6.1.79) gives

$$s_{j+1}^T W s_{j+1} = s_j^T W s_j + 2\alpha_j s_j^T W d_j + \alpha_j^2 d_j^T C d_j \geq s_j^T W s_j \quad (6.1.81)$$

which shows $\|s_j\|_W$ is strictly increasing.

If $s = s_{j+1}$, then (6.1.77) follows directly. If the algorithm stops because $d_j^T B d_j \leq 0$ or $\|s_{j+1}\|_W \geq \Delta$, then the final iterate s is chosen on the boundary, i.e., $\|s\|_W = \Delta$, which is the largest possible length any iterate can have. Therefore (6.1.77) is satisfied. $\square$

Steihaug-CG method is used in Step 3 in Algorithm 6.1.1 for solving the trust-region subproblem. The trust-region method with Steihaug-CG technique is very useful for large-scale optimization problems.

About other techniques of solving subproblems, please consult Gay [144], Moré and Sorensen [222], and Rendl and Wolkowicz [286].

6.2 Conic Model and Collinear Scaling Algorithm

6.2.1 Conic Model

The well-known quadratic model usually considered is

$$q(d) = f(x_k) + g_k^T d + \frac{1}{2} d^T B_k d, \quad (6.2.1)$$

where $g_k = \nabla f(x_k)$ and B_k is a symmetric matrix that is intended to approximate the Hessian matrix. The model (6.2.1) satisfies

$$q(0) = f(x_k), \quad \nabla q(0) = \nabla f(x_k). \quad (6.2.2)$$

In quasi-Newton method, the updates satisfy the quasi-Newton condition

$$B_k(x_k - x_{k-1}) = \nabla f(x_k) - \nabla f(x_{k-1}), \quad (6.2.3)$$

which is just the interpolation condition

$$\nabla q(-d) = \nabla f(x_{k-1}). \quad (6.2.4)$$

Therefore, a secant method based on a quadratic model satisfies the three interpolation conditions in (6.2.2) and (6.2.4). However, a quadratic function simply does not possess enough degrees of freedom to incorporate all of the information in the iterative procedure. It often leads to poor prediction of minimizer by these methods based on a quadratic model, especially for those functions with strong non-quadratic behavior or severely changed curvature.

Davidon [82] proposed a new class of algorithm which is able to interpolate richer information on functions and gradients. Such a model function is more general than the quadratic model. This new model is called a conic model. The new algorithm is called a conic model algorithm or a collinear scaling algorithm.

A smooth function is said to be conic if and only if it is a ratio of a quadratic function to the square of an affine function.

Now, we consider the conic model function

$$c(d) = f(x_k) + \frac{g_k^T d}{1 + b^T d} + \frac{1}{2} \frac{d^T A_k d}{(1 + b^T d)^2}. \quad (6.2.5)$$

Its gradient is

$$\begin{aligned}
 \nabla c(d) &= \frac{(1 + b^T d)g_k - g_k^T db}{(1 + b^T d)^2} + \frac{(1 + b^T d)^2 A_k d - (1 + b^T d)d^T A_k db}{(1 + b^T d)^4} \\
 &= \frac{(1 + b^T d)I - bd^T}{1 + b^T d} \cdot \frac{(1 + b^T d)g_k + A_k d}{(1 + b^T d)^2} \\
 &= \frac{1}{1 + b^T d} \left[I - \frac{bd^T}{1 + b^T d} \right] \left[g_k + \frac{A_k d}{1 + b^T d} \right]. \tag{6.2.6}
 \end{aligned}$$

This gradient vanishes, $\nabla c(d) = 0$, if and only if

$$g_k + \frac{A_k d}{1 + b^T d} = 0. \tag{6.2.7}$$

In this time, the conic model $c(d)$ has minimizer which is by (6.2.7) that

$$d = \frac{-A_k^{-1} g_k}{1 + b^T A_k^{-1} g_k}. \tag{6.2.8}$$

Hence, if $1 + b^T A_k^{-1} g_k \neq 0$, then the desired minimizer is

$$x_{k+1} = x_k - \frac{A_k^{-1} g_k}{1 + b^T A_k^{-1} g_k}. \tag{6.2.9}$$

In fact, an essential ingredient of a conic model is to construct a collinear scaling

$$x(d) - x \triangleq \tilde{d} = \frac{d}{1 + b^T d} \tag{6.2.10}$$

or

$$d = \frac{\tilde{d}}{1 - b^T \tilde{d}}. \tag{6.2.11}$$

In new variable $\tilde{d}$ -space, the conic model (6.2.5) becomes a quadratic model

$$c(\tilde{d}) = f(x_k) + g_k^T \tilde{d} + \frac{1}{2} \tilde{d}^T A_k \tilde{d}. \tag{6.2.12}$$

6.2.2 Generalized Quasi-Newton Equation

By means of collinear scaling, Sorensen [315] derived the generalized quasi-Newton equations that the conic model method satisfies.

Let collinear scaling be

$$x(w) = x + \frac{Jw}{1 + h^T w}, \quad (6.2.13)$$

where $J \in R^{n \times m}$, $h \in R^m$, and $w \in R^m$. The local quadratic model to the scaled objective function $\phi(w) = f(x(w))$ has the form

$$\psi(w) = \phi(0) + \phi'(0)w + \frac{1}{2}w^T Bw. \quad (6.2.14)$$

Obviously,

$$\phi'(w) = f'(x(w))x'(w) \quad (6.2.15)$$

with

$$x'(w) = \frac{1}{1 + h^T w} J \left[I - \frac{wh^T}{1 + h^T w} \right]. \quad (6.2.16)$$

In terms of the objective function f and the matrix J in the collinear scaling, the quadratic model has the form

$$\psi(w) = f(x) + f'(x)Jw + \frac{1}{2}w^T Bw. \quad (6.2.17)$$

If B is positive definite, then the step v that solves

$$v^T B = -f'(x)J \quad (6.2.18)$$

is a predicted minimizer of the scaled function $\phi(w)$. The step s from x to $\bar{x}$ is

$$s = \frac{Jv}{1 + h^T v}, \quad (6.2.19)$$

so that $\bar{x} = x(v)$ in (6.2.13).

Next, we develop the generalized quasi-Newton equations that the conic model satisfies. Let $\bar{x}$ be not an acceptable approximation to a local minimizer of f . Then we wish to update the collinear scaling and also the quadratic model of the new scaled function

$$\bar{\phi}(w) = f(\bar{x}(w)). \quad (6.2.20)$$

Here

$$\bar{x}(w) = \bar{x} + \frac{\bar{J}w}{1 + \bar{h}^T w} \quad (6.2.21)$$

is a collinear scaling with barred quantities $\bar{J}, \bar{h}, \bar{x}$ replacing J, h, x in (6.2.13). The corresponding new quadratic model of the new scaled function is

$$\bar{\psi}(w) = \bar{\phi}(0) + \bar{\phi}'(0)w + \frac{1}{2}w^T \bar{B}w. \quad (6.2.22)$$

For convenience of discussion, we write the derivatives as follows:

$$\bar{x}'(w) = \frac{(1 + \bar{h}^T w)\bar{J} - \bar{J}w\bar{h}^T}{(1 + \bar{h}^T w)^2}, \quad (6.2.23)$$

$$\bar{x}'(0) = \bar{J}, \quad (6.2.24)$$

$$\bar{x}'(-v) = \frac{(1 - \bar{h}^T v)\bar{J} + \bar{J}v\bar{h}^T}{(1 - \bar{h}^T v)^2}, \quad (6.2.25)$$

$$\bar{\phi}'(w) = f'(\bar{x}(w))\bar{x}'(w), \quad (6.2.26)$$

$$\bar{\phi}'(0) = f'(\bar{x}(0))\bar{x}'(0) = f'(\bar{x})\bar{J}, \quad (6.2.27)$$

$$\bar{\phi}'(-v) = f'(\bar{x}(-v))\bar{x}'(-v) = f'(x)(\bar{J} + s\bar{h}^T)/\gamma, \quad (6.2.28)$$

$$\bar{\psi}'(w) = \bar{\phi}'(0) + w^T \bar{B}, \quad (6.2.29)$$

$$\bar{\psi}'(0) = \bar{\phi}'(0) = f'(\bar{x})\bar{J}, \quad (6.2.30)$$

$$\bar{\psi}'(-v) = \bar{\phi}'(0) - v^T \bar{B} = f'(\bar{x})\bar{J} - v^T \bar{B}, \quad (6.2.31)$$

where

$$\gamma = 1 + \bar{h}^T(-v) = 1 - \bar{h}^T v. \quad (6.2.32)$$

Then, (6.2.22) can be written as

$$\bar{\psi}(w) = f(\bar{x}) + \frac{\nabla f(\bar{x})^T \bar{s}}{1 - \bar{h}^T \bar{J}^{-1} \bar{s}} + \frac{1}{2} \frac{\bar{s}^T \bar{J}^{-T} \bar{B} \bar{J}^{-1} \bar{s}}{(1 - \bar{h}^T \bar{J}^{-1} \bar{s})^2} \quad (6.2.33)$$

$$= f(\bar{x}) + f'(\bar{x})\bar{J}w + \frac{1}{2}w^T \bar{B}w, \quad (6.2.34)$$

where

$$\bar{s} = \bar{J}w/(1 + \bar{h}^T w), \quad (6.2.35)$$

$$w = \bar{J}^{-1} \bar{s}/(1 - \bar{h}^T \bar{J}^{-1} \bar{s}). \quad (6.2.36)$$

To update J, h, B to $\bar{J}, \bar{h}, \bar{B}$, we introduce the consistency condition

$$\bar{x}(0) = \bar{x}, \quad \bar{x}(-v) = x \quad (6.2.37)$$

and the interpolation conditions:

$$\bar{\psi}(0) = \bar{\phi}(0), \quad \bar{\psi}'(0) = \bar{\phi}'(0), \quad (6.2.38)$$

$$\bar{\psi}(-v) = \bar{\phi}(-v), \quad \bar{\psi}'(-v) = \bar{\phi}'(-v). \quad (6.2.39)$$

From the consistency condition $\bar{x}(-v) = x$, we have

$$x = \bar{x}(-v) = \bar{x} - \bar{J}v/\gamma, \quad (6.2.40)$$

that is

$$\bar{J}v = \gamma s, \quad (6.2.41)$$

where $s = \bar{x} - x$. Obviously, conditions (6.2.38) are immediately met by the quadratic model (6.2.22). Also, consider the interpolation conditions (6.2.39); since

$$\begin{aligned} \bar{\psi}(-v) &= \bar{\phi}(0) - \bar{\phi}'(0)v + \frac{1}{2}v^T \bar{B}v \\ &= f(\bar{x}) - f'(\bar{x})\bar{J}v + \frac{1}{2}v^T \bar{B}v \\ &= f(\bar{x}) - \gamma f'(\bar{x})s + \frac{1}{2}v^T \bar{B}v \end{aligned}$$

and

$$\bar{\phi}(-v) = f(x),$$

then the first equation of (6.2.39) becomes

$$f(x) = f(\bar{x}) - \gamma f'(\bar{x})s + \frac{1}{2}v^T \bar{B}v. \quad (6.2.42)$$

Similarly, it follows from (6.2.31) and (6.2.28) that the second equation of (6.2.39) becomes

$$f'(x)(\bar{J} + s\bar{h}^T)/\gamma = f'(\bar{x})\bar{J} - v^T \bar{B}, \quad (6.2.43)$$

that can be written as

$$\bar{B}v = r, \quad (6.2.44)$$

where

$$r^T = \bar{\phi}'(0) - \bar{\phi}'(-v) = f'(\bar{x})\bar{J} - f'(x)(\bar{J} + s\bar{h}^T)/\gamma, \quad (6.2.45)$$

which is the gradient difference of the scaled function.

Then, we obtain a generalized quasi-Newton equation

$$\bar{B}v = r, \quad \bar{J}v = \gamma s, \quad \bar{h}^T v = 1 - \gamma, \quad (6.2.46)$$

where r is defined by (6.2.45). In particular, when $\bar{J} = I, \bar{h} = 0, \gamma = 1$, the generalized quasi-Newton equations reduce to the usual quasi-Newton equation

$$\bar{B}v = r. \quad (6.2.47)$$

At this time, $v = s = \bar{x} - x$ and $r = f'(\bar{x}) - f'(x)$.

It remains to determine the choices of γ . By the second and the third equations of (6.2.46), we have

$$(\bar{J} + s\bar{h}^T)v = s, \quad (6.2.48)$$

so that

$$v^T \bar{B}v = r^T v = (\gamma f'(\bar{x}) - f'(x)/\gamma)s \triangleq y^T s, \quad (6.2.49)$$

where

$$y = \gamma f'(\bar{x})^T - f'(x)^T/\gamma. \quad (6.2.50)$$

Substituting the above into (6.2.42) gives

$$\gamma^2 f'(\bar{x})s + 2\gamma[f(x) - f(\bar{x})] + f'(x)s = 0. \quad (6.2.51)$$

To make γ real, we must require

$$\rho^2 \triangleq (f(\bar{x}) - f(x))^2 - (f'(\bar{x})s)(f'(x)s) \geq 0. \quad (6.2.52)$$

If $\bar{B}$ is to be positive definite, then we obtain

$$v^T \bar{B}v = 2\rho \quad (6.2.53)$$

from (6.2.49) by taking

$$\gamma = \frac{-f'(x)s}{f(x) - f(\bar{x}) + \rho} \quad (6.2.54)$$

$$= \frac{f(x) - f(\bar{x}) + \rho}{-f'(\bar{x})s} \quad (6.2.55)$$

as the positive root of (6.2.51).

For the one-dimensional case, the corresponding conic model iteration is as follows.

Algorithm 6.2.1 (*Conic Model Algorithm for One-dimensional Case*)

Step 0. Given x_1, s_1 , evaluate f_1, f'_1 at x_1 ;

Step k. for $k = 1, 2, \dots$

Step k.1 set $x_{k+1} = x_k + s_k$;

Step k.2 evaluate f_{k+1}, f'_{k+1} ;

Step k.3 set $\rho_k = ((f_k - f_{k+1})^2 - (f'_k s_k)(f'_{k+1} s_k))^{\frac{1}{2}}$;
 $\gamma_k = -f'_k s_k / (f_k - f_{k+1} + \rho_k)$;

Step k.4 $s_{k+1} = s_k / [(1/\gamma_k^3)(f'_k/f'_{k+1}) - 1]$. $\square$

6.2.3 Updates that Preserve Past Information

Based on the generalized quasi-Newton equations and other criteria, we can obtain some updates about J, h , and B .

Let $\mathcal{W}$ be the linear span of previous scaled search directions and let $\bar{\mathcal{W}} = \text{span}\{\mathcal{W}, v\}$. Then a natural requirement is that

$$\bar{\phi}(w - v) = \phi(w), \quad \forall w \in N_0 \subset \mathcal{W}, \quad (6.2.56)$$

where $N_0 = \{w \in \mathcal{W} : 1 + h^T w > 0\}$. Condition (6.2.56) immediately leads to the requirement

$$\bar{x}(w - v) = x(w), \quad \forall w \in N_0 \subset \mathcal{W}. \quad (6.2.57)$$

Since $\bar{x}(-v) = x$ and $\bar{x}(0) = \bar{x}$, it follows that

$$\begin{aligned} \bar{x}(w - v) &= \bar{x}(0) + \frac{\bar{J}(w - v)}{\bar{h}^T(w - v) + 1} \\ &= x + \frac{\bar{J}v}{\gamma} + \frac{\bar{J}(w - v)}{\bar{h}^T w + \gamma} \quad (\text{by (6.2.46)(iii)}) \\ &= x + \frac{\bar{J}v(\bar{h}^T w / \gamma) + \bar{J}w}{\bar{h}^T w + \gamma} \\ &= x + \frac{(\bar{J} + s\bar{h}^T)w}{\bar{h}^T w + \gamma} \quad (\text{by (6.2.46)(ii)}). \end{aligned} \quad (6.2.58)$$

By (6.2.57) and (6.2.58), we have

$$x + \frac{(\bar{J} + s\bar{h}^T)w}{\bar{h}^T w + \gamma} = x(w) = x + \frac{Jw}{h^T w + 1}. \quad (6.2.59)$$

Set $w = \alpha p, p \in N_0 \subset \mathcal{W}, \alpha \in [0, 1]$. Matching the coefficients of α on both sides of (6.2.59) yields

$$(\bar{J} + s\bar{h}^T)p = \gamma Jp, \quad \bar{h}^T p = \gamma h^T p$$

for every $p \in N_0$. Then we obtain

$$(\bar{J} + s\bar{h}^T)w = \gamma Jw \quad (6.2.60)$$

and

$$\bar{h}^T w = \gamma h^T w, \quad \forall w \in \mathcal{W}. \quad (6.2.61)$$

Since

$$s = \frac{Jv}{h^T v + 1},$$

then (6.2.60) becomes

$$(\bar{J} + \gamma s h^T)w = \gamma Jw, \quad \forall w \in \mathcal{W},$$

that is

$$\bar{J} = \gamma(J - s h^T) \quad (6.2.62)$$

satisfying $\bar{J}v = \gamma s$ as well as (6.2.60). The equation (6.2.62) is an update about J .

Next, we discuss the update about h . Note that $\bar{h}$ satisfies

$$\bar{h}^T w = \gamma h^T w, \quad \bar{h}^T v = 1 - \gamma. \quad (6.2.63)$$

Now let Q be an orthogonal projector on $\mathcal{W}$ and $P = I - Q$. Let

$$\bar{h} = Qc + Pd, \quad (6.2.64)$$

where c and d are arbitrary vectors. Multiplying (6.2.64) by w^T gives

$$\gamma h^T w = \bar{h}^T w = w^T Qc = c^T w,$$

then we take $c = \gamma h$. Further, multiplying (6.2.64) by v^T yields

$$1 - \gamma = \bar{h}^T v = \gamma v^T Qh + v^T Pd. \quad (6.2.65)$$

Then

$$\bar{h}^T v = 1 - \gamma = \gamma v^T Q h + \frac{1 - \gamma - \gamma v^T Q h}{v^T P d} v^T P d. \quad (6.2.66)$$

Hence, we take

$$\bar{h} = \gamma Q h + P d \quad (6.2.67)$$

or

$$\bar{h} = \gamma Q h + \frac{1 - \gamma - \gamma v^T Q h}{v^T P d} P d \quad (6.2.68)$$

as long as $v^T P d \neq 0$. So, we obtain the updates about h .

By use of (6.2.27) and (6.2.28), it follows from (6.2.60) that

$$\begin{aligned} \bar{\phi}'(-v)w &= \frac{f'(x)}{\gamma}(\bar{J} + s\bar{h}^T)w \\ &= \frac{f'(x)}{\gamma} \cdot \gamma J w \\ &= f'(x)J w \\ &= \phi'(0)w. \end{aligned} \quad (6.2.69)$$

To update the Hessian of the quadratic model of a scaled function, the following requirements are imposed:

$$\bar{\psi}(w - v) = \psi(w), \quad (6.2.70)$$

$$\bar{\psi}'(w - v)q = \psi'(w)q, \quad (6.2.71)$$

for all $w, q \in \mathcal{W}$. Condition (6.2.70) implies that

$$\bar{\phi}(0) + \bar{\phi}'(0)(w - v) + \frac{1}{2}(w - v)^T \bar{B}(w - v) = \phi(0) + \phi'(0)w + \frac{1}{2}w^T B w$$

for all $w \in \mathcal{W}$. Arranging it gives

$$\begin{aligned} &\left[\bar{\phi}(0) - \bar{\phi}'(0)v + \frac{1}{2}v^T \bar{B}v - \phi(0) \right] + [\bar{\phi}'(0) - \phi'(0) - v^T \bar{B}]w \\ &+ \frac{1}{2}w^T (\bar{B} - B)w = 0, \quad \forall w \in \mathcal{W}. \end{aligned}$$

The first term vanishes identically due to (6.2.42), and the second term vanishes due to (6.2.60) and (6.2.43). Therefore, we have

$$w^T (\bar{B} - B)w = 0, \quad \forall w \in \mathcal{W}. \quad (6.2.72)$$

Similarly, condition (6.2.71) implies

$$[\bar{\phi}'(0) + (w - v)^T \bar{B}]q = [\phi'(0) + w^T B]q,$$

that is

$$[\bar{\phi}'(0) - \phi'(0) - v^T \bar{B}]q + w^T (\bar{B} - B)q = 0, \forall w, q \in \mathcal{W}.$$

The first bracket in the left-hand side of the above equation vanishes due to (6.2.60) and (6.2.43). Then we also get

$$w^T (\bar{B} - B)q = 0, \forall w, q \in \mathcal{W}. \quad (6.2.73)$$

Hence, the above discussion shows that if and only if

$$w^T (\bar{B} - B)q = 0, \forall w, q \in \mathcal{W}, \quad (6.2.74)$$

both (6.2.70) and (6.2.71) are satisfied.

Consequently, the required update satisfies

$$\bar{B}v = r, w^T (\bar{B} - B)q = 0, \forall w, q \in \mathcal{W}. \quad (6.2.75)$$

The above can be written as

$$\begin{aligned} \bar{B} &= \mathcal{U}_Q(B, v, r) \\ &= \left\{ \bar{B} \left| \begin{array}{l} \bar{B}v = r, Q^T (\bar{B} - B)Q = 0, \bar{B} \text{ symmetric,} \\ Q \text{ is an orthogonal projector in } \mathcal{W}. \end{array} \right. \right\}. \end{aligned} \quad (6.2.76)$$

Here the update class coming from additional requirements (6.2.70)-(6.2.71) is bigger than the update class due to Schnabel [299]:

$$\{\bar{B} \mid \bar{B}v = r, (\bar{B} - B)w = 0, \forall w \in \mathcal{W}, \bar{B} \text{ symmetric}\}.$$

Also, the update (6.2.75) includes the optimal conditioning update due to Davidon [80].

From the above discussion, we have obtained a class of updates:

$$\bar{J} = \gamma(J - sh^T), \quad (6.2.77)$$

$$\bar{h} = \gamma Qh + \frac{1 - \gamma - \gamma v^T Qh}{v^T P d} P d, \quad (6.2.78)$$

$$w^T (\bar{B} - B)q = 0, \forall w, q \in \mathcal{W}. \quad (6.2.79)$$

6.2.4 Collinear Scaling BFGS Algorithm

Sorensen [315] developed a collinear scaling BFGS algorithm without projections. That is, in (6.2.77)–(6.2.79) we take $P = I$ and $Q = 0$. Further, we take

$$d = \bar{J}^T g, \quad s = \bar{x} - x, \quad y = \gamma \bar{g} - g/\gamma.$$

Then we obtain the following updating formulas:

$$\bar{J} = \gamma(J - sh^T), \quad (6.2.80)$$

$$\bar{h} = \left(\frac{1 - \gamma}{\gamma g^T s} \right) \bar{J}^T g, \quad (6.2.81)$$

$$\bar{H} = H + \frac{v(v - Hr)^T}{v^T r} + \frac{(v - Hr)v^T}{v^T r} - \frac{r^T(v - Hr)}{(v^T r)^2} vv^T, \quad (6.2.82)$$

which is called a collinear scaling BFGS formula for updating the inverse Hessian approximation H , where $H = B^{-1}$ and $\bar{H} = \bar{B}^{-1}$.

Further, denote

$$C = JHJ^T, \quad \bar{C} = \bar{J}\bar{H}\bar{J}^T. \quad (6.2.83)$$

Using (6.2.80), (6.2.81) and (6.2.45), we have

$$\begin{aligned} r &= \bar{J}^T \bar{g} - \frac{1}{\gamma}(\bar{J}^T + \bar{h}s^T)g = \bar{J}^T \bar{g} - \frac{1}{\gamma}(\bar{J}^T + \frac{1 - \gamma}{\gamma} \bar{J}^T)g \\ &= \bar{J}^T \bar{g} - \frac{1}{\gamma^2} \bar{J}^T g = (J - sh^T)^T y. \end{aligned} \quad (6.2.84)$$

Thus

$$\bar{J} \left(I - \frac{vr^T}{v^T r} \right) = \gamma(J - sh^T) - \frac{\gamma s(J^T y - hs^T y)^T}{s^T y} = \gamma \left(I - \frac{sy^T}{s^T y} \right) J. \quad (6.2.85)$$

Equation (6.2.85) allows us to obtain

$$\bar{C} = \gamma^2 \left[\left(I - \frac{sy^T}{s^T y} \right) C \left(I - \frac{ys^T}{s^T y} \right) + \frac{ss^T}{s^T y} \right]. \quad (6.2.86)$$

So, instead of updating J and H , we only need to update C .

By (6.2.21), the scaled direction is

$$s_{k+1} = \frac{1}{1 + h_{k+1}^T v_{k+1}} J_{k+1} v_{k+1} \quad (\text{note } v_{k+1} = -H_{k+1} J_{k+1}^T g_{k+1})$$

$$\begin{aligned}
 &= \frac{-J_{k+1}H_{k+1}J_{k+1}^T g_{k+1}}{1 - (1 - \gamma_k)g_k^T J_{k+1}H_{k+1}J_{k+1}^T g_{k+1}/(\gamma_k g_k^T s_k)} \\
 &\triangleq \frac{-J_{k+1}H_{k+1}J_{k+1}^T g_{k+1}}{1 + \delta_{k+1}} \\
 &\triangleq -\theta_{k+1}C_{k+1}g_{k+1}.
 \end{aligned} \tag{6.2.87}$$

Hence, we obtain the iterative scheme

$$x_{k+1} = x_k - \theta_k C_k g_k, \tag{6.2.88}$$

where

$$\begin{aligned}
 \theta_k &= 1/(1 + \delta_k), \\
 \delta_k &= -(1 - \gamma_{k-1})g_{k-1}^T C_k g_k / (\gamma_{k-1}g_{k-1}^T s_{k-1}), \\
 C_k &= \mathcal{U}(C_{k-1}, s_{k-1}, y_{k-1}).
 \end{aligned}$$

In the following, we give a description of the algorithm.

Algorithm 6.2.2 (*Collinear Scaling BFGS Algorithm*)

Step 1. Initialize C_0 positive definite, $x_0, \delta_0, \alpha_{\max} > 0$. Compute f_0, g_0 . Set $k = 0$.

Step 2. If $\delta_k < 0$, set $\bar{\alpha} = \min(\alpha_{\max}, -1/\delta_k)$, else $\bar{\alpha} := \alpha_{\max}$. Do line search for the function

$$\phi(\alpha) \triangleq f\left(x_k - \frac{\alpha}{1 + \alpha\delta_k} C_k g_k\right)$$

and find $\alpha_k \in (0, \bar{\alpha})$. Set

$$\begin{aligned}
 s_k &= -\frac{\alpha_k}{1 + \alpha_k\delta_k} C_k g_k, \\
 x_{k+1} &= x_k + s_k, \\
 f_{k+1} &= f(x_{k+1}), \quad g_{k+1}^T = f'(x_{k+1}), \\
 \rho^2 &= (f_k - f_{k+1})^2 - (g_{k+1}^T s_k)(g_k^T s_k),
 \end{aligned}$$

such that $\rho^2 > 0$ and $f_{k+1} < f_k$.

Step 3. If “convergence” then stop.

Step 4. Compute

$$\begin{aligned}\gamma_k &= -g_k^T s_k / (f_k - f_{k+1} + \rho), \quad y_k = \gamma_k g_{k+1} - g_k / \gamma_k, \\ C_{k+1} &= \gamma_k^2 [(I - s_k y_k^T / s_k^T y_k) C_k (I - y_k s_k^T / s_k^T y_k) + s_k s_k^T / s_k^T y_k], \\ \delta_{k+1} &= -(1 - \gamma_k) g_k^T C_{k+1} g_{k+1} / \gamma_k g_k^T s_k.\end{aligned}$$

Set $k := k + 1$, go to Step 2. $\square$

Following the Broyden-Dennis-Moré convergence theory about quasi-Newton methods, we can establish Q-superlinear convergence of the collinear scaling BFGS algorithm, i.e.,

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0.$$

Furthermore, Di and Sun [101] propose a conic trust-region method for unconstrained optimization.

Let x denote the current approximation of the minimizer and let

$$f = f(x), \quad g = g(x) = \nabla f(x). \quad (6.2.89)$$

Then the conic trust-region model of $f(x + s)$ is

$$\min \quad \psi(s) = f + \frac{g^T s}{1 - a^T s} + \frac{1}{2} \frac{s^T A s}{(1 - a^T s)^2}, \quad (6.2.90)$$

$$\text{s.t.} \quad \|Ds\| \leq \Delta, \quad (6.2.91)$$

where $A \in R^{n \times n}$ is the Hessian approximation at x , $a \in R^n$ is a horizontal vector such that $1 - a^T s > 0$, D is a scaling matrix and Δ a trust-region radius. The above subproblem can be written as

$$\min \quad f + g^T Jw + \frac{1}{2} w^T Bw \quad (6.2.92)$$

$$\text{s.t.} \quad s = Jw / (1 + h^T w), \quad \|Ds\| \leq \Delta. \quad (6.2.93)$$

Di and Sun [101] discussed the necessary and sufficient condition of the solution for the conic trust-region subproblem, presented an algorithm and established the global and superlinear convergence. Besides, Zhu etc. [384] discussed a quasi-Newton type trust-region method based on a conic model for solving unconstrained optimization. Sun and Yuan [337], Sun, Yuan, and Yuan [338] studied a conic trust-region algorithm for linear and nonlinear constrained optimization respectively. About the topic of conic model method, readers are referred also to Grandinetti [162], Ariyawansa [3], Sun [333], Sheng [308], Han, Sun et al. [168].

6.3 Tensor Methods

The tensor method is also a generalization of the quadratic model method. In fact, the tensor method is based on the third- or fourth-order model for optimization problems, and intends to improve upon the efficiency and reliability of standard methods on problem where $\nabla^2 f(x^*)$ is singular.

The tensor method was introduced by Schnabel and Frank [302] for solving systems of nonlinear equations and by Schnabel and Chow [301] for unconstrained optimization, respectively. In this section, we will describe the tensor methods for nonlinear equations and for unconstrained optimization.

6.3.1 Tensor Method for Nonlinear Equations

Let $F : R^n \rightarrow R^n$. Consider solving nonlinear equations

$$F(x) = 0, \quad (6.3.1)$$

that is to find $x^* \in R^n$ so that $F(x^*) = 0$. Newton's method for (6.3.1) is defined as

$$x_+ = x_c - F'(x_c)^{-1}F(x_c), \quad (6.3.2)$$

when $F'(x_c)$ is nonsingular, where x_c and x_+ denote the current and the next iterate respectively. Newton's method is based on the linear model at x_c ,

$$M(x_c + d) = F(x_c) + F'(x_c)d. \quad (6.3.3)$$

As we know, the outstanding advantage of Newton's method is its rapid convergence, that is if $F'(x_c)$ is Lipschitz continuous in the neighborhood of x^* and $F'(x^*)$ is nonsingular, then the sequence produced by (6.3.2) locally and quadratically converges to x^* . This implies that there are $\delta > 0$ and $c \geq 0$, such that when $\|x_0 - x^*\| \leq \delta$, the iterative sequence $\{x_k\}$ satisfies

$$\|x_{k+1} - x^*\| \leq c\|x_k - x^*\|^2. \quad (6.3.4)$$

However, if $F'(x^*)$ is singular, then the iterative sequence does not converge rapidly. The tensor method described in this section will overcome the shortcoming, and we can see that the tensor method still has rapid convergence when $F'(x^*)$ is singular.

Consider the second-order model

$$M_T(x_c + d) = F(x_c) + F'(x_c)d + \frac{1}{2}T_cdd, \quad (6.3.5)$$

where $T_c \in R^{n \times n \times n}$ is a three-dimensional tensor. Usually, the (6.3.5) is said to be a tensor model, and the corresponding method is called a tensor method. To discuss the tensor method, we first give a definition concerning these tensors.

Definition 6.3.1 *Let $T \in R^{n \times n \times n}$. Then T consists of n horizontal faces $H_i \in R^{n \times n}$, $i = 1, \dots, n$, where $H_i[j, k] = T[i, j, k]$. For $v, w \in R^n$, we have $Tvw \in R^n$ with the i -th component*

$$Tvw[i] = v^T H_i w = \sum_{j=1}^n \sum_{k=1}^n T[i, j, k] v[j] w[k]. \quad (6.3.6)$$

Hence, the tensor model given in (6.3.5) is, in fact, an n -dimensional vector in which each component is a quadratic model of the component of $F(x)$, i.e.,

$$(M_T(x_c + d))[i] = f_i + g_i^T d + \frac{1}{2} d^T H_i d, \quad i = 1, \dots, n, \quad (6.3.7)$$

where $f_i = F(x_c)[i]$, g_i^T is the i -th row of $F'(x_c)$, and H_i the Hessian matrix of the i -th component of $F(x)$.

An obvious choice of T_c in (6.3.5) is $F''(x_c)$. However, the computational amount is prohibitive, since, in each iteration, it needs to compute n^3 second-order partial derivatives of $F''(x_c)$, store over $n^3/2$ elements, and solve n quadratic equations in n unknowns. To overcome these drawbacks, the tensor method constructs T_c in low-rank by using available information of function values and first derivatives. So, the additional efforts are small related to the standard method.

To construct T_c , we select p not necessarily consecutive past iterates $x_{-1}, \dots, x_{-p}$ and ask the model (6.3.5) to interpolate $F(x)$ at these points, i.e.,

$$F(x_{-k}) = F(x_c) + F'(x_c)s_k + \frac{1}{2} T_c s_k s_k, \quad k = 1, \dots, p, \quad (6.3.8)$$

where

$$s_k = x_{-k} - x_c, \quad k = 1, \dots, p. \quad (6.3.9)$$

The selected directions $\{s_k\}$ are required to be strongly independent, i.e., make the angle between each direction s_k and the subspace spanned by other directions have at least θ degree. Values of θ between 20 and 40 degrees have proven to be best in practice. This procedure is easily implemented by

using a modified Gram-Schmidt algorithm. Since directions $\{s_k\}$ are linearly independent, then $p \leq n$. In practice, one takes

$$p \leq \sqrt{n}.$$

Now we write (6.3.8) as

$$T_c s_k s_k = z_k, \quad k = 1, \dots, p, \quad (6.3.10)$$

where

$$z_k = 2(F(x_{-k}) - F(x_c) - F'(x_c)s_k). \quad (6.3.11)$$

The (6.3.10) is a set of $np \leq n^{3/2}$ linear equations in n^3 unknowns $T_c[i, j, k]$, $1 \leq i, j, k \leq n$. We choose the smallest symmetric T_c , in the Frobenius norm, which satisfies the equations (6.3.10). Below, we choose T_c following the technique for a secant update with the smallest change in quasi-Newton methods (see Chapter 5).

First, we define the three-dimensional rank-one tensor.

Definition 6.3.2 Let $u, v, w \in R^n$. The tensor $T \in R^{n \times n \times n}$, for which

$$T[i, j, k] = u[i] \cdot v[j] \cdot w[k], \quad (1 \leq i, j, k \leq n), \quad (6.3.12)$$

is called a third-order rank-one tensor of $T \in R^{n \times n \times n}$ and is denoted by

$$T = u \otimes v \otimes w. \quad (6.3.13)$$

Obviously, the i -th horizontal face of the rank-one tensor $u \otimes v \otimes w$ is a rank-one matrix $u[i](vw^T)$.

Theorem 6.3.3 Let $p \leq n$. Let $s_k \in R^n$, $k = 1, \dots, p$ with $\{s_k\}$ linearly independent, and let $z_k \in R^n$, $k = 1, \dots, p$. Define $M \in R^{p \times p}$ by $M[i, j] = (s_i^T s_j)^2$, $1 \leq i, j \leq p$, and define $Z \in R^{n \times p}$ with z_k the k -th column, $k = 1, \dots, p$. Then M is positive definite, and the solution to

$$\min_{T_c \in R^{n \times n \times n}} \|T_c\|_F \quad (6.3.14)$$

$$\text{s.t.} \quad T_c s_k s_k = z_k, \quad k = 1, \dots, p \quad (6.3.15)$$

is

$$T_c = \sum_{k=1}^p (a_k \otimes s_k \otimes s_k), \quad (6.3.16)$$

where a_k is the k -th column of $A \in R^{n \times p}$ and $A = M^{-1}Z$.

Proof. Since the objective function and constraints can be decomposed into n separate objective functions and constraints, then (6.3.14)-(6.3.15) are equivalent to the following separate minimization problem

$$\min_{H_i \in R^{n \times n}} \|H_i\|_F \quad (6.3.17)$$

$$\text{s.t.} \quad s_k^T H_i s_k = z_k[i], \quad k = 1, \dots, p, \quad (6.3.18)$$

where H_i are the horizontal faces of T_c , $i = 1, \dots, n$. Note that the problem (6.3.17)-(6.3.18) is a sub-determined set of p equations in n^2 unknowns.

Let $h_i \in R^{n^2}$,

$$\begin{aligned} h_i = & (H_i[1, 1], H_i[1, 2], \dots, H_i[1, n], H_i[2, 1], \dots, \\ & H_i[2, n], \dots, H_i[n, 1], \dots, H_i[n, n])^T. \end{aligned} \quad (6.3.19)$$

Let $\bar{S} \in R^{p \times n^2}$, the k -th row of $\bar{S}$ is

$$\bar{s}_k = (s_k[1]s_k^T, s_k[2]s_k^T, \dots, s_k[n]s_k^T). \quad (6.3.20)$$

Let also the i -th row of $Z \in R^{n \times p}$ be $\bar{z}_i$,

$$\bar{z}_i \in R^p, \quad \bar{z}_i[k] = z_k[i], \quad 1 \leq i \leq n, \quad 1 \leq k \leq p.$$

Then (6.3.17) is equivalent to

$$\min_{h_i \in R^{n^2}} \|h_i\|_2 \quad (6.3.21)$$

$$\text{s.t.} \quad \bar{S}h_i = \bar{z}_i^T. \quad (6.3.22)$$

Note that the $\{s_k\}$ are linearly independent, then $\bar{S}$ is full row rank, and hence the solution to (6.3.21)-(6.3.22) is

$$h_i = \bar{S}^T (\bar{S} \bar{S}^T)^{-1} \bar{z}_i^T. \quad (6.3.23)$$

Since $M = \bar{S} \bar{S}^T$, then M is positive definite. Also,

$$\begin{bmatrix} \bar{a}_1 \\ \vdots \\ \bar{a}_n \end{bmatrix} = A = M^{-1}Z = (\bar{S} \bar{S}^T)^{-1} \begin{bmatrix} \bar{z}_1 \\ \vdots \\ \bar{z}_n \end{bmatrix}. \quad (6.3.24)$$

Hence the i -th row of A is

$$\bar{a}_i = (\bar{S} \bar{S}_i^T)^{-1} \bar{z}_i. \quad (6.3.25)$$

Therefore (6.3.23) means

$$h_i = \bar{S}^T \bar{a}_i^T. \quad (6.3.26)$$

Note that here $\bar{a}_i$ is the i -th row of A and a_k is the k -th column of A , then

$$\bar{a}_i[k] = a_k[i], \quad 1 \leq i \leq n, \quad 1 \leq k \leq p.$$

Then it follows from (6.3.26) that

$$h_i = \sum_{k=1}^p \bar{a}_i[k] \bar{s}_k^T = \sum_{k=1}^p a_k[i] \bar{s}_k^T, \quad (6.3.27)$$

where $\bar{s}_k$ is defined by (6.3.20) and the k -th row of $\bar{S}$, the $\bar{s}_k^T$ denotes a transpose of $\bar{s}_k$ and a column vector with n^2 elements.

Returning to (6.3.27) in the terms of H_i and s_k , and using (6.3.19) and (6.3.20) give

$$H_i = \sum_{k=1}^p a_k[i] s_k s_k^T. \quad (6.3.28)$$

Finally, combining n matrices H_i gives the desired T_c in (6.3.16).

Substituting (6.3.16) into (6.3.5), the tensor model has the form

$$M_T(x_c + d) = F(x_c) + F'(x_c)d + \frac{1}{2} \sum_{k=1}^p a_k (d^T s_k)^2. \quad (6.3.29)$$

In the above model, the simple form of the second-order term is a key to efficiently find a minimizer of this model. In the tensor method, the additional $4pn$ storage are required to save $\{a_k\}, \{s_k\}, \{x_{-k}\}$ and $\{F(x_{-k})\}$. Additional cost is $n^2p + O(np^2)$ operations for computing $A = M^{-1}Z$. Since $p \leq \sqrt{n}$, this is a very small additional cost, more than the cost of the standard quadratic model method.

6.3.2 Tensor Methods for Unconstrained Optimization

In this subsection, we extend the tensor method to solving unconstrained optimization problem

$$\min_{x \in R^n} f(x), \quad f: R^n \rightarrow R. \quad (6.3.30)$$

Note that the standard quadratic model methods do not converge quickly if the Hessian $\nabla^2 f(x^*)$ is singular. In this case, the convergence rate is linear

at best. Furthermore, the third derivatives do not supply information in the direction where the second derivative matrix is lacking. Thus, adding an approximation to $\nabla^3 f(x_c)$ alone will not lead to better-than-linear convergence. Therefore, we consider employing the following fourth order tensor model

$$m_T(x_c + d) = f(x_c) + \nabla f(x_c) \cdot d + \frac{1}{2} \nabla^2 f(x_c) \cdot d^2 + \frac{1}{6} T_c \cdot d^3 + \frac{1}{24} V_c \cdot d^4, \quad (6.3.31)$$

where $T_c \in R^{n \times n \times n}$, a three-dimensional tensor and $V_c \in R^{n \times n \times n \times n}$, a four-dimensional tensor, are symmetric. Equation (6.3.31) is called a tensor model for unconstrained optimization; the methods based on (6.3.31) are referred to tensor methods.

How to Choose T_c and V_c ?

To select T_c and V_c , we select p not necessarily consecutive past iterates $x_{-1}, \dots, x_{-p}$, and ask that the model (6.3.31) interpolate $f(x)$ and $\nabla f(x)$ at these points, i.e.,

$$\begin{aligned} f(x_{-k}) &= f(x_c) + \nabla f(x_c) \cdot s_k + \frac{1}{2} \nabla^2 f(x_c) \cdot s_k^2 + \frac{1}{6} T_c \cdot s_k^3 \\ &\quad + \frac{1}{24} V_c \cdot s_k^4, \end{aligned} \quad (6.3.32)$$

$$\begin{aligned} \nabla f(x_{-k}) &= \nabla f(x_c) + \nabla^2 f(x_c) \cdot s_k + \frac{1}{2} T_c \cdot s_k^2 \\ &\quad + \frac{1}{6} V_c \cdot s_k^3, \end{aligned} \quad (6.3.33)$$

where $s_k = x_{-k} - x_c$, $k = 1, \dots, p$. As in the previous subsection, the directions $\{s_k\}$ are strongly linearly independent. We also set $p \leq n^{1/3}$.

Multiplying (6.3.33) by s_k gives

$$\nabla f(x_{-k}) \cdot s_k = \nabla f(x_c) \cdot s_k + \nabla^2 f(x_c) \cdot s_k^2 + \frac{1}{2} T_c \cdot s_k^3 + \frac{1}{6} V_c \cdot s_k^4. \quad (6.3.34)$$

Define $\alpha, \beta \in R^p$ respectively by

$$\alpha[k] = T_c \cdot s_k^3, \quad (6.3.35)$$

$$\beta[k] = V_c \cdot s_k^4, \quad (6.3.36)$$

where $k = 1, \dots, p$. Then (6.3.34) and (6.3.32) have the following form respectively:

$$\frac{1}{2} \alpha[k] + \frac{1}{6} \beta[k] = q_1[k], \quad (6.3.37)$$

$$\frac{1}{6}\alpha[k] + \frac{1}{24}\beta[k] = q_2[k], \quad (6.3.38)$$

where

$$q_1[k] = \nabla f(x_{-k}) \cdot s_k - \nabla f(x_c) \cdot s_k - \nabla^2 f(x_c) \cdot s_k^2, \quad (6.3.39)$$

$$q_2[k] = f(x_{-k}) - f(x_c) - \nabla f(x_c) \cdot s_k - \frac{1}{2}\nabla^2 f(x_c) \cdot s_k^2, \quad (6.3.40)$$

for $k = 1, 2, \dots, p$. The system (6.3.37)-(6.3.38) is nonsingular, so each $\alpha[k]$ and $\beta[k]$ are uniquely determined. Thus, we can determine V_c by the minimization problem

$$\begin{aligned} \min_{V_c \in R^{n \times n \times n \times n}} \quad & \|V_c\|_F \\ \text{s.t.} \quad & V_c \cdot s_k^4 = \beta[k], \quad k = 1, \dots, p. \\ & V_c \text{ symmetric.} \end{aligned} \quad (6.3.41)$$

We then substitute the obtained value of V_c into (6.3.33), obtaining

$$T_c \cdot s_k^2 = a_k, \quad k = 1, \dots, p, \quad (6.3.42)$$

where

$$a_k = 2 \left(\nabla f(x_{-k}) - \nabla f(x_c) - \nabla^2 f(x_c) \cdot s_k - \frac{1}{6}V \cdot s_k^3 \right).$$

This is a set of $np \leq n^{4/3}$ linear equations in n^3 unknowns $T_c[i, j, k]$, $1 \leq i, j, k \leq n$. Then we determine T_c by the minimization problem

$$\begin{aligned} \min_{T_c \in R^{n \times n \times n}} \quad & \|T_c\|_F \\ \text{s.t.} \quad & T_c \cdot s_i^2 = a_i, \quad i = 1, \dots, p \\ & T_c \text{ symmetric.} \end{aligned} \quad (6.3.43)$$

The following two theorems give the solutions of problems (6.3.41) and (6.3.43).

Theorem 6.3.4 *Let $p \leq n$. Let $s_k \in R^n$, $k = 1, \dots, p$ with $\{s_k\}$ linearly independent, and let $\beta \in R^p$. Define $M \in R^{p \times p}$ by $M[i, j] = (s_i^T s_j)^4$, $1 \leq i, j \leq p$. Define $\gamma \in R^p$ by $\gamma = M^{-1}\beta$. Then the solution to (6.3.41) is*

$$V_c = \sum_{k=1}^p \gamma[k] (s_k \otimes s_k \otimes s_k \otimes s_k). \quad (6.3.44)$$

Proof. Define $\hat{v} \in R^{n^4}$ by

$$\hat{v}^T = (V_c[1, 1, 1, 1], V_c[1, 1, 1, 2], \dots, V_c[1, 1, 1, n], \\ V_c[1, 1, 2, 1], \dots, V_c[1, 1, 2, n], \dots, V_c[n, n, n, n]).$$

Let the matrix $\hat{S} \in R^{p \times n^4}$ with the k -th row as

$$(s_k[1])^4, (s_k[1])^3(s_k[2]), \dots, (s_k[1])^3(s_k[n]), \dots, (s_k[n])^4.$$

Then, (6.3.41) is equivalent to

$$\min_{\hat{v}} \quad \|\hat{v}\|_2 \quad (6.3.45)$$

$$\text{s.t.} \quad \hat{S}\hat{v} = \beta, \quad V_c \text{ symmetric}, \quad (6.3.46)$$

where V_c is the original form of $\hat{v}$. Since $\{s_k\}$ are linearly independent, $\hat{S}$ has full row rank. Hence, the solution to

$$\min_{\hat{v}} \quad \|\hat{v}\|_2 \quad (6.3.47)$$

$$\text{s.t.} \quad \hat{S}\hat{v} = \beta \quad (6.3.48)$$

is

$$\hat{v} = \hat{S}^T(\hat{S}\hat{S}^T)^{-1}\beta = \hat{S}^T M^{-1}\beta = \hat{S}^T \gamma, \quad (6.3.49)$$

where $M = \hat{S}\hat{S}^T$. By reversing the transformation from $\hat{v}$ to V_c , we get (6.3.44). Since V_c is symmetric, it is the solution of (6.3.41). $\square$

Theorem 6.3.5 *Let $p \leq n$. Let $s_k \in R^n$, $k = 1, \dots, p$ with $\{s_k\}$ linearly independent, and let $a_k \in R^n$, $k = 1, \dots, p$. Then the solution to problem (6.3.43) is*

$$T_c = \sum_{k=1}^p (b_k \otimes s_k \otimes s_k + s_k \otimes b_k \otimes s_k + s_k \otimes s_k \otimes b_k), \quad (6.3.50)$$

where $b_k \in R^n$, $k = 1, \dots, p$, and $\{b_k\}$ is the unique set of vectors for which (6.3.50) satisfies

$$T_c s_i^2 = a_i, \quad i = 1, \dots, p.$$

Proof. First, we show that the constraint set in (6.3.43) is feasible. Let $t_i \in R^n$, $i = 1, \dots, p$ satisfy

$$t_i^T s_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad \text{for } j = 1, \dots, p.$$

Since $\{s_i\}$ are linearly independent, such vectors t_i can be obtained via a QR factorization. Then

$$T = \sum_{i=1}^p (t_i \otimes t_i \otimes a_i + t_i \otimes a_i \otimes t_i + a_i \otimes t_i \otimes t_i - 2(a_i^T s_i)(t_i \otimes t_i \otimes t_i))$$

is a feasible solution to (6.3.43).

Dennis and Schnabel [93] assume that the set of tensors $T_j \in R^{n \times n \times n}$ is generated by the following procedure: $T_0 = 0$ and for $j = 0, 1, \dots$, T_{2j+1} is the solution of

$$\min \|T_{2j+1} - T_{2j}\|_F \quad (6.3.51)$$

$$\text{s.t. } T_{2j+1} \cdot s_i^2 = a_i, \quad i = 1, \dots, p, \quad (6.3.52)$$

and T_{2j+2} is the solution of

$$\min \|T_{2j+2} - T_{2j+1}\|_F \quad (6.3.53)$$

$$\text{s.t. } T_{2j+2} \text{ symmetric.} \quad (6.3.54)$$

Then the sequence $\{T_j\}$ has a limit which is the unique solution to (6.3.43). (see the derivation of Powell symmetric Broyden update in §5.1).

Next, we show that this limit has form (6.3.50) for some set of vectors $\{b_k\}$, by showing that each T_{2j} has this form.

Trivially, it is true for T_0 . Assume it is true for some j , i.e.,

$$T_{2j} = \sum_{k=1}^p (u_k \otimes s_k \otimes s_k + s_k \otimes u_k \otimes s_k + s_k \otimes s_k \otimes u_k) \quad (6.3.55)$$

for some set of vectors u_k . Then from Theorem 6.3.3, the solution to (6.3.51)-(6.3.52) is

$$T_{2j+1} = T_{2j} + \sum_{k=1}^p (v_k \otimes s_k \otimes s_k)$$

for some set of vectors $\{v_k\}$. Thus

$$\begin{aligned} T_{2j+2} &= T_{2j} + \frac{1}{3} \sum_{k=1}^p (v_k \otimes s_k \otimes s_k + s_k \otimes v_k \otimes s_k + s_k \otimes s_k \otimes v_k) \\ &= \sum_{k=1}^p \left(\left(u_k + \frac{v_k}{3} \right) \otimes s_k \otimes s_k + s_k \otimes \left(u_k + \frac{v_k}{3} \right) \otimes s_k \right. \\ &\quad \left. + s_k \otimes s_k \otimes \left(u_k + \frac{v_k}{3} \right) \right), \end{aligned} \quad (6.3.56)$$

which again has the form (6.3.55). Thus, by induction, the solution T_c must have the form (6.3.50) for some set of vectors $\{b_k\}$.

Finally, we show that the set of vectors $\{b_k\}$, for which T_c given by (6.3.50) satisfies

$$T_c s_i^2 = a_i, \quad i = 1, \dots, p, \quad (6.3.57)$$

is unique. This will mean that equations (6.3.50) and (6.3.57) uniquely determine the solution to (6.3.43). In fact, substituting (6.3.50) into (6.3.57) gives a system of np linear equations in np unknowns, where the matrix is a function of $\{s_k\}$, the unknowns are the elements of the $\{b_k\}$, and the right-hand side consists of the elements of the $\{a_k\}$.

Since we have showed above that (6.3.43) is feasible for any $\{a_k\}$, the above derivation and the theory of Dennis-Schnabel [93] imply that for any set $\{s_k\}$, this linear system has at least one solution for any right-hand side. Therefore, the linear system must be nonsingular and have a unique solution. This means that the set of vectors $\{b_k\}$ is uniquely determined. $\square$

Solving the Tensor Model

Substituting the values of T_c and V_c in (6.3.50) and (6.3.44) into the tensor model (6.3.31) gives

$$\begin{aligned} m_T(x_c + d) &= f(x_c) + \nabla f(x_c) \cdot d + \frac{1}{2} \nabla^2 f(x_c) \cdot d^2 \\ &\quad + \frac{1}{2} \sum_{k=1}^p (b_k^T d) (s_k^T d)^2 + \frac{1}{24} \sum_{k=1}^p \gamma[k] (s_k^T d)^4 \\ &= f(x_c) + g^T d + \frac{1}{2} d^T H d \\ &\quad + \frac{1}{2} \sum_{k=1}^p (b_k^T d) (s_k^T d)^2 + \frac{1}{24} \sum_{k=1}^p \gamma[k] (s_k^T d)^4, \end{aligned} \quad (6.3.58)$$

where $g = \nabla f(x_c)$, $H = \nabla^2 f(x_c)$.

Let $S \in R^{n \times p}$ with k -th column s_k . Let $Z \in R^{n \times (n-p)}$ and $W \in R^{n \times p}$ have full column rank and satisfy $Z^T S = 0$ and $W^T S = I$, respectively. The Z and W can be calculated through the QR factorization of S .

Write

$$d = Wu + Zt, \quad (6.3.59)$$

where $u \in R^p$ and $t \in R^{n-p}$. Substituting (6.3.59) into (6.3.58) gives

$$m_T(x_c + Wu + Zt) = f(x_c) + g^T Wu + g^T Zt + \frac{1}{2} u^T W^T H W u$$

$$\begin{aligned}
& +u^T W^T H Z t + \frac{1}{2} t^T Z^T H Z t + \frac{1}{2} \sum_{k=1}^p u[k]^2 (b_k^T W u + b_k^T Z t) \\
& + \frac{1}{24} \sum_{k=1}^p \gamma[k] u[k]^4,
\end{aligned} \tag{6.3.60}$$

which is a quadratic with respect to t . Therefore, for the tensor model to have a minimizer, $Z^T H Z$ must be positive definite and the derivative of the model with respect to t must be 0, i.e.,

$$Z^T g + Z^T H Z t + Z^T H W u + \frac{1}{2} Z^T \sum_{i=1}^p b_i u[i]^2 = 0. \tag{6.3.61}$$

Therefore

$$t = -(Z^T H Z)^{-1} Z^T \left(g + H W u + \frac{1}{2} \sum_{i=1}^p b_i u[i]^2 \right). \tag{6.3.62}$$

Substituting (6.3.62) into (6.3.60) reduces the problem of minimizing the tensor model to finding a minimizer of

$$\begin{aligned}
\hat{m}_T(u) &= f + g^T W u + \frac{1}{2} u^T W^T H W u + \frac{1}{2} \sum_{i=1}^p u[i]^2 (b_i^T W u) \\
&+ \frac{1}{24} \sum_{i=1}^p \gamma[i] u[i]^4 - \frac{1}{2} \left(g + H W u + \frac{1}{2} \sum_{i=1}^p b_i u[i]^2 \right)^T \\
&\cdot Z (Z^T H Z)^{-1} Z^T \left(g + H W u + \frac{1}{2} \sum_{i=1}^p b_i u[i]^2 \right),
\end{aligned} \tag{6.3.63}$$

which is a fourth-degree polynomial in u -variable. If (6.3.63) has a minimizer u^* , then the minimizer of the original tensor model (6.3.58) is given by

$$d^* = W u^* + Z t^*, \tag{6.3.64}$$

where t^* is determined by setting $u = u^*$ in (6.3.60).

In implementation we may employ line search or trust-region strategy. If the obtained direction d^* is a descent direction, but $x_c + d^*$ is not acceptable, we set $x_+ = x_c + \lambda d^*$ where λ is a steplength factor. If (6.3.63) has no minimizer, or d^* is not in a descent direction, we find the next iterate by using a line search algorithm based on the standard quadratic model.

Similarly, we can also use trust-region technique studied in §6.1. The trust-region tensor model is

$$\min_{d \in R^n} m_T(x_c + d) \quad (6.3.65)$$

$$\text{s.t.} \quad \|d\|_2 \leq \Delta_c, \quad (6.3.66)$$

where $\Delta_c \in R$ is the trust-region radius.

The tensor algorithm for unconstrained optimization is as follows.

Algorithm 6.3.6 (*Tensor Method*) Given $x_c, f(x_c), \Delta_c$.

Step 1. Calculate $\nabla f(x_c)$, and decide whether to stop. If not, go to Step 2.

Step 2. Calculate $\nabla^2 f(x_c)$.

Step 3. Select p past points from among the $n^{1/3}$ most recent past points.

Step 4. Calculate T_c and V_c .

Step 5. Find a potential acceptable next iterate $x_c + d_T$ and a potential new trust-region radius Δ_T by using the tensor model and a trust-region technique.

Step 6. Find a potential acceptable next iterate $x_c + d_N$ and a potential new trust-region radius Δ_N by using the quadratic model and a trust-region technique.

Step 7. If $f(x_c + d_T) \leq f(x_c + d_N)$, then set

$$x_+ = x_c + d_T, \Delta_+ = \Delta_T;$$

else set

$$x_+ = x_c + d_N, \Delta_+ = \Delta_N.$$

Step 8. Set $x_c = x_+$, $f(x_c) = f(x_+)$, $\Delta_c = \Delta_+$, and go to Step 1.
□

Note that in the tensor method, the Hessian can be replaced by finite difference Hessian approximation or secant updates, because the cost of computing a Hessian is large. Also, we would like to point out that the tensor

method is a generalization of the standard quadratic model method. However, there are still various problems waiting for us to solve. For example, the strategy of computing both tensor model and quadratic model at each iteration is not ideal; how to choose a suitable p , how to use the tensor method in constrained problems and so on. This kind of method is worth doing further study.

Exercises

1. Let $f(x) = x_1^4 + x_1^2 + x_2^2$. Let the current iterate $x^{(k)} = (1, 1)^T$, $\Delta_k = \frac{1}{2}$. Try using double-dogleg method to find $x^{(k+1)}$.

2. Let $f(x) = \frac{1}{2}x_1^2 + x_2^2$. Let the starting point $x^{(0)} = (1, 1)^T$. For $\Delta_0 = 1$ and $\Delta_0 = \frac{5}{4}$,

(1) Use dogleg method to find $x^{(1)}$.

(2) Use double-dogleg method to find $x^{(2)}$.

3. Let s_k be an approximate solution of subproblem (6.1.1). Show that s_k satisfies

$$q^{(k)}(0) - q^{(k)}(s_k) \geq \beta \|g_k\|_2 \min \left\{ \Delta_k, \frac{\|g_k\|_2}{\|B_k\|_2} \right\},$$

where $\beta \in (0, 1]$.

4. What is the attractive point of the trust-region method?

5. Use trust-region Newton method to minimize the Rosenbrock function (see Appendix: Problem 1.1).

6. Use trust-region quasi-Newton method to minimize the extended Rosenbrock function (see Appendix: Problem 1.2).

7. Consider using dogleg method to construct path $s(\tau)$. Show that $\|s(\tau)\|$ increases monotonically along this path.

8. Derive expression (6.1.30)–(6.1.31) of the Cauchy point.

9. (1) Let s_k^G solve

$$\begin{aligned} \min \quad & f(x_k) + g_k^T s \\ \text{s.t.} \quad & \|Ds\| \leq \Delta_k. \end{aligned}$$

Show that

$$s_k^G = -\frac{\Delta_k}{\|D^{-1}g_k\|_2} D^{-2} g_k.$$

(2) The generalized Cauchy point can be defined by

$$q^{(k)}(s_k^c) = \min\{q^{(k)}(s) \mid s = \tau s_k^G, \|Ds\| \leq \Delta_k\},$$

where s_k^G is defined by (1). Therefore, the generalized Cauchy point can be expressed as

$$s_k^c = \tau_k s_k^G = -\tau_k \frac{\Delta_k}{\|D^{-1}g_k\|_2} D^{-2} g_k, \quad (6.3.67)$$

where

$$\tau_k = \arg \min_{\tau > 0} q^{(k)}(\tau s_k^G) \quad \text{s.t.} \quad \|\tau D s_k^G\| \leq \Delta_k.$$

Show:

$$\tau_k = \begin{cases} 1 & \text{if } g_k^T D^{-2} B_k D^{-2} g_k \leq 0; \\ \min\{\|D^{-1}g_k\|_2^3 / (\Delta_k g_k^T D^{-2} B_k D^{-2} g_k), 1\} & \text{otherwise.} \end{cases} \quad (6.3.68)$$

10. Mimic Theorem 6.1.2, state and prove the necessary and sufficient condition that s^* is the solution of subproblem

$$\begin{aligned} \min_s \quad & f + g^T s + \frac{1}{2} s^T B s \\ \text{s.t.} \quad & \|Ds\|_2 \leq \Delta. \end{aligned}$$

11. Write out the program of Steihaug-CG Algorithm 6.1.12.

12. Try to state the relations among quadratic model, conic model, tensor model and collinear scaling.

13. Starting from collinear scaling $s = \frac{w}{1+h^T w}$, derive a generalized quasi-Newton equation.
14. Derive the collinear scaling BFGS formula. Try to derive other formulas of collinear scaling.